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Master Thesis

Hardware Implementation of Gate-Current Optimization Techniques for Switching Loss and Electromagnetic Emission Reduction in SiC MOSFETs

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Abstract

In modern power electronic applications, efficient switching of power MOSFETs is critical, requiring simultaneous minimization of switching losses and electromagnetic emissions (EME). However, conventional voltage-source gate drivers employing fixed gate resistances (R_G) offer limited flexibility in gate-current control, resulting in suboptimal switching waveforms that impose a strict trade-off between switching loss reduction and EME suppression. This limitation becomes increasingly critical for wide-bandgap devices such as silicon carbide (SiC) MOSFETs, which exhibit fast switching transitions. This thesis focuses on the practical realization of a gate-current optimization framework aimed at simultaneously minimizing switching energy and EME in real hardware implementations. Initially, the theoretical foundations of switching loss and EME in power MOSFETs are established. Subsequently, the state-space simulation framework employed in this work is described and adapted for a Bosch SiC MOSFET by incorporating experimentally extracted device characteristics and parasitic parameters. Following that, the definitions and formulas of the switching loss and EME-related metrics (Heisenberg-Gabor Inequality (HGI)) used in this work are introduced. Baseline simulations using fixed gate resistances are performed to quantify the inherent limitations of conventional gate-driving methods with respect to switching loss and EME reduction. Thereafter, a gate-current optimization framework based on a genetic algorithm is integrated into the simulation environment. The optimization employs a multi-objective cost function that quantitatively balances switching loss and EME while constraining the drain-source voltage (V_{DS}) overshoot within specified limits. The key contribution of this thesis lies in the experimental realization of these optimized gate-current profiles using a programmable current-source gate driver (EG120). Detailed experimental validation is conducted on a custom-built double pulse testing board, during which the hardware was operated at peak drain currents of up to 275 A and peak drain–source voltages of up to 1000 V. The experimental results of the optimized gate-current profile approach demonstrate significant improvements over conventional fixed gate-resistance methods. The loss-optimized EG120 configuration shows a reduction in the low-side device switching loss by 14.4% (vs. 5.1 Ω R_G case) and 42.6% (vs. 7.5 Ω R_G case), while simultaneously lowering the high-side device HGI by 21.7% and the V_{DS} overshoot by 47.4%. Furthermore, the EME-optimized EG120 configuration achieves a 58.6% reduction in the high-side HGI and a 59.7% reduction in V_{DS} overshoot compared to the fixed 5.1 Ω R_G baseline. Observed discrepancies between simulated and measured results are systematically analyzed, highlighting the limitations of the current source driver employed. Overall, the results of this work demonstrate the strong potential of active gate drivers, capable of delivering tailored gate-current profiles. By demonstrating the practical feasibility of optimized gate-current shaping in real hardware, this thesis bridges the gap between theoretical optimization and experimental realization, enabling improved efficiency and EME behavior in power electronic converters.

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1 Introduction

1.1 Background

In modern power electronics, the power MOSFET is increasingly becoming a key enabling device for global electrification [1]. The present market landscape demands high-efficiency power conversion in electric vehicle (EV) powertrains, battery management systems (BMS), and on-board chargers (OBCs). Furthermore, the rapid growth of renewable energy systems, such as solar inverters and wind turbines, imposes stringent requirements on power semiconductor devices to possess high blocking voltage capability, while requiring power converters to achieve high power density and operate at higher switching speeds [1]. These requirements are met by the ongoing transition from bipolar silicon (Si) technologies (e.g., insulated-gate bipolar transistors (IGBTs)) to unipolar wide-bandgap (WBG) devices, such as silicon carbide (SiC) and gallium nitride (GaN) MOSFETs. However, the resulting steep voltage and current transitions associated with faster switching significantly increase electromagnetic emissions (EME) from the device and subsequently might cause electromagnetic interference (EMI) in nearby circuits [2]. This necessitates careful design considerations at the device, packaging, and system levels.

The efficiency of a power converter is largely dictated by the commutation strategy employed during transitions between the conducting (ON) and blocking (OFF) states of the power device. In hard-switching operation, the MOSFET undergoes switching transition while both the drain-source voltage (V_{DS}) and the drain current (I_D) are non-zero, resulting in an overlap of voltage and current during the switching interval. This overlap produces instantaneous power dissipation [3]. In contrast, soft-switching techniques such as zero-voltage switching (ZVS) and zero-current switching (ZCS) enforce either the voltage or the current to reach zero before the switching transition, thereby minimizing this overlap and reducing switching losses [4].

To achieve soft switching operation, auxiliary hardware such as active clamping circuits or zero voltage transition (ZVT) networks is often employed [4]. However, the integration of these additional resonant components can increase the converter size, cost, and control complexity, while simultaneously introducing new potential failure modes that can degrade system reliability. Alternatively, dead time optimization enables partial or full ZVS without additional components by utilizing the device's intrinsic output capacitance (C_{oss}) to discharge the drain-source voltage in advance during a carefully timed dead-time between switching states [5]. Yet, applying this technique to traction inverters proves challenging due to the wide variation in load current, operating voltage, and junction temperature during normal operation. This variability makes fixed dead time settings inherently suboptimal, while adaptive approaches require precise timing resolution on the order of nanoseconds, significantly increasing system complexity [5]. As a consequence, hard-switching operation remains the dominant approach in most traction inverter applications [6].

1.2 Motivation

In hard-switching topologies, converter efficiency is intrinsically linked to the commutation speed of the power semiconductor device. Fast switching transitions, characterized by high rates of change of the drain-source voltage ($\mathrm{d}v_{DS}/\mathrm{d}t$) and drain current ($\mathrm{d}i_D/\mathrm{d}t$), are essential for minimizing the voltage-current crossover duration, thereby reducing switching losses [3]. However, these rapid transitions excite parasitic inductances and capacitances, generating high-frequency ringing and noise that result in significant EME.

To comply with EMC standards, this high-frequency content is often suppressed by slowing the switching transitions using a larger external gate resistance (R_G), among other mitigation techniques [7, 8]. However, this method extends the overlap between voltage and current, directly increasing switching losses. This conflict highlights the limitations of conventional voltage source gate drivers (VSGDs). By relying on fixed gate resistances, VSGDs impose a strict compromise between system efficiency and regulatory compliance [9].

Consequently, this thesis proposes a transition toward Current Source Gate Driving (CSGD) strategies. Unlike VSGDs, current-source driver topologies provide the capability to actively control the gate charge delivery, offering a theoretical pathway to reduce the dependence of EME generation on switching speed [10] [11]. Building on the theoretical switching loss–EME optimization framework presented in [12], this thesis incorporates the practical constraints of the EG120 CSGD into the optimization process. The primary focus is on the practical realization of the resulting optimized gate-current profiles in hardware, while addressing the key challenges associated with translating theoretically optimal solutions into real-world power electronic systems.

1.3 Research Questions

This thesis investigates the following research questions:

1. *Optimal switching definition:* How does existing literature define an “optimal” hard switching event in terms of switching losses and EME, and what gate-current profile characteristics (shape and magnitude) are required to realize it?
2. *Simulation and optimization:* How can existing optimization algorithms be employed and tuned within a simulation environment to derive gate-current waveforms that approach the optimal switching behavior?
3. *Practical implementation:* How can such theoretically optimized gate-current profiles be applied to a MOSFET in a practical hardware realization?
4. *Implementation limitations:* What technical constraints and challenges arise when implementing these gate-current profiles in real-world power electronic systems?

1.4 Approach and Structure of the Work

This thesis is structured as follows.

Chapter 2 presents the theoretical foundations of switching losses and EME in power MOSFETs. It begins with an overview of ideal and real switching behavior, followed by the description of switching losses. The chapter then discusses the origins of EME in power MOSFET-based converters and introduces a figure of merit (FOM) used to characterize these emissions.

Chapter 3 introduces the state-space model used for simulation of the power MOSFET. The chapter details the device and parasitic modeling approach, the derivation of device voltages and currents, and the performance calculation methodology.

Chapter 4 describes the gate-current optimization framework. This chapter outlines the employed weight functions, the different genetic algorithm-based optimization strategies, and presents simulation results obtained using optimized gate-current profiles.

Chapter 5 details the experimental realization and validation of the optimized gate-current profiles on a custom-built double-pulse test (DPT) hardware. It presents a comprehensive analysis of the experimental results, validating the effectiveness of the optimized profiles in reducing switching losses and EME compared to fixed gate-resistance baselines. Furthermore, discrepancies between simulation and measurement are analyzed to identify the practical limitations of the implemented current-source gate driver.

2 Theory of Switching Loss and Electromagnetic Emissions in Power MOSFETs

2.1 Hard-Switching Operation of SiC MOSFETs

2.1.1 Double Pulse Test

The double-pulse test (DPT) is a widely used experimental technique for characterizing the dynamic switching behavior of power MOSFETs under well-defined operating conditions [13]. In a typical DPT setup, the device under test is operated in a half-bridge configuration with an inductive load. In this thesis, the low-side (LS) switch is considered the device under test, while the high-side (HS) switch acts as a passive freewheeling device (Fig. 2.1).

Two consecutive gate pulses are applied to the LS switch. The first pulse establishes a defined load current by energizing the inductive load, and the associated turn-off transition is used to evaluate the turn-off switching behavior. The second pulse then forces a controlled turn-on event at the pre-established current level, enabling detailed analysis of the turn-on switching transient (Fig. 2.1). By separating the current buildup from the actual switching event, the DPT allows precise measurement of switching times, voltage and current overshoots, switching energy, and diode reverse-recovery effects [13].

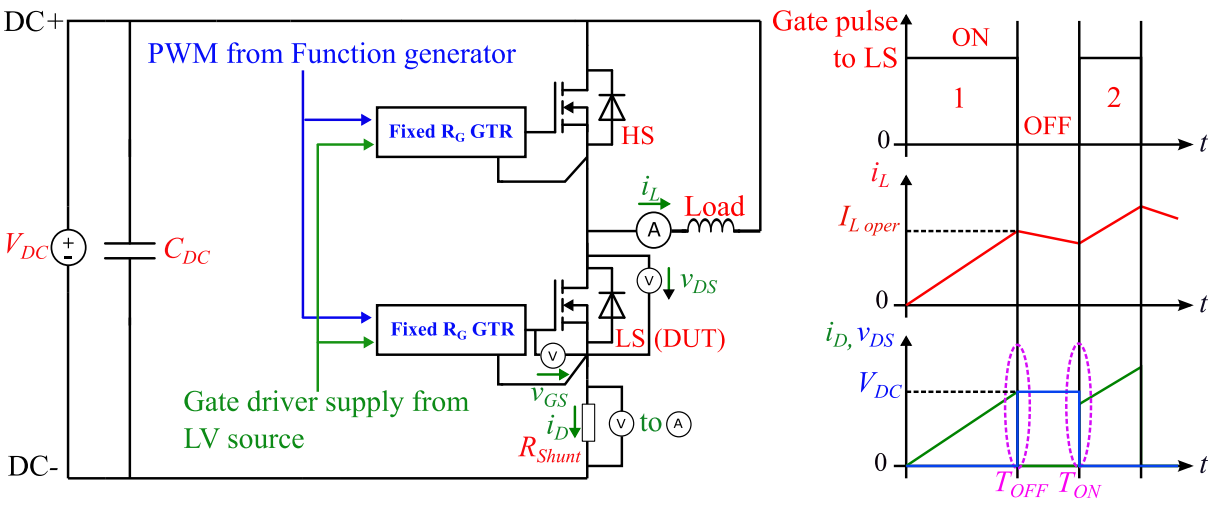


Figure 2.1: Double Pulse Test setup and operation.

2.1.2 Ideal Switching Behavior

In the ideal case, the MOSFET is assumed to switch in the absence of parasitic effects caused by nonlinear device capacitances and parasitic inductances. Under these simplifying assumptions, the switching event can be illustrated using a simplified representation, as shown in Fig. 2.2.

2.1.2.1 Ideal Active Turn-On Transition

- **For** $t < t_1$ **(Off-State):** The gate-source voltage (v_{GS}) is below the threshold voltage $V_{GS,\mathrm{th}}$. The device blocks the full DC-link voltage ($v_{DS} = V_{DC}$), and the drain current i_D is zero.
- **For** $t_1 < t < t_2$ **(Current Rise):** At t_1, V_{GS} exceeds the $V_{GS,\mathrm{th}}$. The drain current i_D rises toward the load current I_L, while the drain-source voltage v_{DS} remains clamped at the DC-link voltage V_{DC}, since part of the load current is still conducted by the freewheeling (body) diode of the high-side (HS) switch (Fig. 2.2).
- **For** $t_2 < t < t_3$ **(Voltage Fall):** Once the current commutation from the HS diode to the LS switch is complete at t_2, the body diode of HS is reverse biased and turns off. v_{DS} then falls approximately linearly toward zero. During this interval, v_{GS} is clamped at the Miller plateau voltage ($V_{GS,\mathrm{plt}}$) due to the charging of the gate-drain capacitance C_{GD} (Fig. 2.2).
- **For** $t > t_3$ **(On-State):** v_{DS} is approximately zero, and v_{GS} rises from the Miller plateau to the full gate driver voltage ($V_{GS,drv}$).

2.1.2.2 Ideal Active Turn-Off Transition

- $t_1 < t < t_2$ **(Voltage Rise):** v_{GS} is discharged to the Miller plateau voltage. During this interval, v_{DS} rises approximately linearly from zero to V_{DC}, while the i_D remains approximately constant at I_L (Fig. 2.2).
- $t_2 < t < t_3$ **(Current Fall):** Once v_{DS} reaches V_{DC}, the freewheeling (body) diode of the high-side (HS) switch becomes forward-biased and takes over the load current. Consequently, i_D of LS decreases approximately linearly to zero.

2.1.3 Real Switching Behavior

In reality, the presence of parasitic capacitances (C_{GS}, C_{GD}, C_{DS}), loop inductance (L_σ), and diode reverse recovery drastically changes the switching profile.

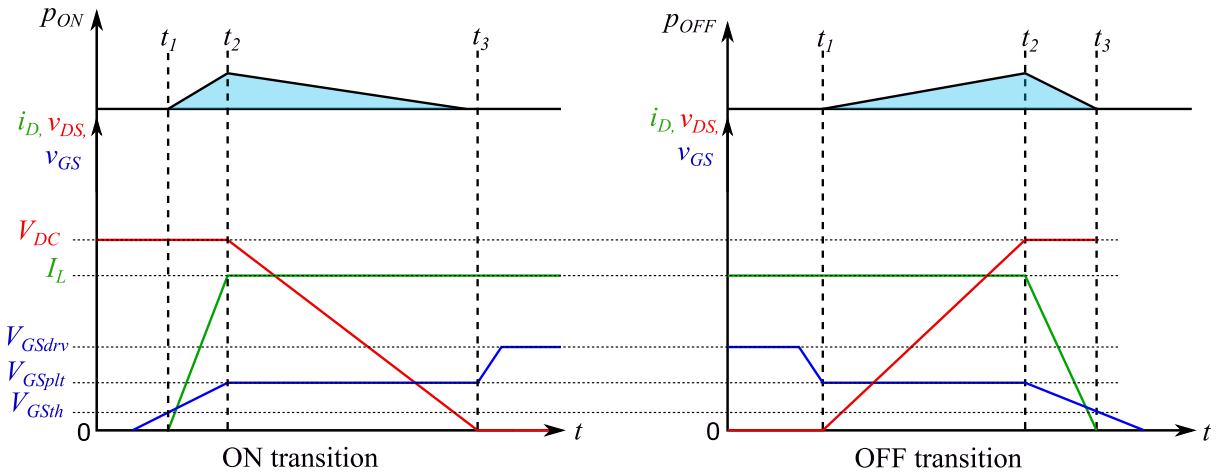


Figure 2.2: Ideal ON and OFF switching transitions.

2.1.3.1 Real Active Turn-On Transition

- $t < t_1$ **(Turn-On Delay):** The gate driver charges the input capacitance C_{iss}, defined as the sum of C_{GS} and C_{GD}. The v_{GS} rises from a negative off-state voltage (e.g., −5 V) to the threshold voltage (e.g., 3 V). The i_D remains at zero.
- $t_1 < t < t_2$ **(Current Commutation):** i_D rises rapidly. The high $\mathrm{d}i_D/\mathrm{d}t$ through L_σ creates a voltage drop, causing v_{DS} to decrease slightly before the main transition [13] (Fig. 2.3):

$$v_{DS} = V_{DC} - L_\sigma \frac{\mathrm{d}i_D}{\mathrm{d}t} \tag{2.1}$$

- $t_2 < t < t_3$ **(i_D peak):** Ideally, current commutation ends at this point. Unlike silicon MOSFETs, the body diode of a SiC MOSFET exhibits negligible stored minority-carrier charge. Consequently, the observed current overshoot is primarily caused by the capacitive displacement current required to charge the device output capacitance, C_{oss} ($C_{\text{oss}} = C_{GD} + C_{DS}$) [14]. During this interval, the capacitive current can be approximated as

$$i_{\text{cap}} \approx C_{\text{oss}}(v_{DS}) \cdot \frac{\mathrm{d}v_{DS}}{\mathrm{d}t} \tag{2.2}$$

$$i_{D,\text{peak}} \approx I_L + i_{\text{cap}} \tag{2.3}$$

- $t_3 < t < t_4$ **(Voltage Fall & Miller Plateau):** Once i_D reaches its peak, the HS diode enters the blocking state, allowing v_{DS} to decrease. The rapid $\mathrm{d}v_{DS}/\mathrm{d}t$ drives a displacement current through C_{GD}, holding v_{GS} at the Miller plateau. The Miller plateau is not perfectly flat but exhibits a positive slope due to the finite transconductance (g_m) and Drain-Induced Barrier Lowering (DIBL). As v_{DS} decreases, the DIBL weakens, causing the effective V_{th} to increase. Consequently, v_{GS} must rise to maintain the constant load current I_L, creating a visible ramp in the plateau [15]. (Fig. 2.3).
- $t > t_4$ **(Post-Charge of v_{GS}):** The v_{DS} reaches its minimum on-state value ($V_{DS,\text{on}}$), and the v_{GS} rises to the full driver voltage.

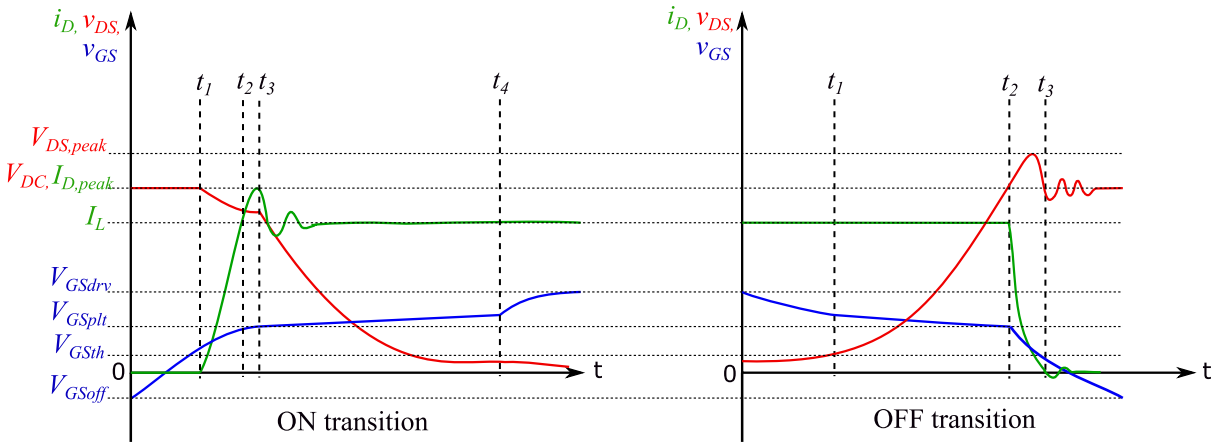


Figure 2.3: Real ON and OFF switching transitions.

2.1.3.2 Real Active Turn-Off Transition

- $t < t_1$ **(Turn-Off Delay):** The driver discharges the gate. As v_{GS} decreases toward the Miller plateau, the channel conductivity is reduced, leading to an increase in $R_{\mathrm{DS,on}}$ and a corresponding rise in v_{DS}, while the drain current i_D remains approximately constant.
- $t_1 < t < t_2$ **(Voltage Rise):** v_{GS} remains near the Miller plateau while v_{DS} rises. During this interval, the gate-driver current is primarily consumed by charging the C_{GD}. However, in SiC MOSFETs the Miller plateau is not perfectly flat and exhibits a slight negative slope mainly due to DIBL. As v_{DS} increases, the DIBL effect reduces the effective threshold voltage, modifying the channel charge required to sustain the load current and causing a small deviation in v_{GS} [15] (Fig. 2.3).
- $t_2 < t < t_3$ **(Current Fall & Voltage Spike):** When v_{DS} reaches V_{DC}, the HS body diode turns on and I_D falls. The negative $\mathrm{d}i_D/\mathrm{d}t$ induces a voltage across L_σ that adds to the DC bus voltage, creating a voltage spike ($V_{DS,\mathrm{peak}}$):

$$V_{DS,peak} = V_{DC} + L_\sigma \cdot \frac{\mathrm{d}i_D}{\mathrm{d}t} \tag{2.4}$$

  The v_{GS} continues to decrease from the Miller plateau below threshold voltage effectively turning the device OFF [16]. The drain current falls to zero and v_{DS} reaches the DC-link voltage.
- $t > t_3$ **(Post discharge of v_{GS}):** The v_{GS} continues to decrease to the final negative off-state voltage (e.g., $V_{GS,off} = -5\,\mathrm{V}$), thereby preventing unintended turn-on caused by parasitic capacitances and other gate coupling mechanisms [17]. Parasitic inductance (L_σ) and output capacitance (C_{oss}) may resonate, causing high-frequency ringing on the v_{DS} waveform (Fig. 2.3).

2.2 Switching Energy

In an ideal switching scenario, v_{DS} and i_{D} overlap to form a triangular instantaneous power waveform [3]. In practice, non-ideal effects such as overshoot, ringing, and plateau regions distort this behavior, rendering triangular approximations inaccurate. Accordingly, the switching

energy can be evaluated by numerically integrating the instantaneous power over the defined switching interval.

The total switching energy, E_{SW}, can defined as the sum of the turn-on and turn-off energy components [3]:

$$E_{\mathrm{on}} = \int_{t_{\mathrm{start,on}}}^{t_{\mathrm{end,on}}} v_{\mathrm{DS}}(t)\, i_{\mathrm{D}}(t)\, \mathrm{d}t \tag{2.5}$$

$$E_{\mathrm{off}} = \int_{t_{\mathrm{start,off}}}^{t_{\mathrm{end,off}}} v_{\mathrm{DS}}(t)\, i_{\mathrm{D}}(t)\, \mathrm{d}t \tag{2.6}$$

The average switching power loss is then given by:

$$P_{\mathrm{SW}} = (E_{\mathrm{on}} + E_{\mathrm{off}})\, f_{\mathrm{sw}} \tag{2.7}$$

Here, $t_{\mathrm{start,on}}$ and $t_{\mathrm{end,on}}$ denote the beginning and end of the turn-on switching interval, while $t_{\mathrm{start,off}}$ and $t_{\mathrm{end,off}}$ denote the corresponding turn-off interval. The exact definition of these intervals depends on the application, switching strategy, and the selected voltage and current threshold criteria.

2.2.1 Methods to Reduce Switching Losses

Switching losses can be reduced by shortening the voltage and current transition intervals (see (2.5) and (2.6)) and by mitigating parasitic effects that distort the switching waveforms.

Reduction of Gate Resistance (R_{G}):

Reducing the external gate resistance increases the gate current, accelerates the charging and discharging of the gate capacitances, and shortens the voltage and current transition intervals, thereby reducing switching losses [2]. However, excessively low gate resistance may excite parasitic oscillations, increase voltage overshoot, and aggravate EME.

Active Gate Driving (AGD):

Active gate driving dynamically adapts the effective driving strength during a switching event to address conflicting requirements at different transition phases. Fast transitions are applied to reduce overlap losses, while slower transitions are used to suppress ringing and EME [18]. Compared to fixed-resistance gate driving, AGD enables improved loss reduction without compromising EMC requirements.

Supporting Techniques:

Additional measures are commonly employed to further reduce switching losses and improve switching robustness:

- **Kelvin source connection:** A dedicated low-inductance gate return path eliminates the negative feedback from the common-source inductance ($L_S \cdot \frac{di_D}{dt}$). Therefore, the driver maintains a higher effective gate current during the switching phase [19]. This shortens the power loss crossover time and thereby directly reduces the switching energy.

- **Negative gate turn-off voltage:** Applying a negative turn-off voltage improves immunity against false turn-on and accelerates gate charge removal [20].
- **Minimization of commutation loop inductance:** Optimized PCB and busbar layouts reduce voltage overshoot and ringing, enabling stable high-speed switching and minimizing associated losses [21].

2.3 Electromagnetic Emissions During Power MOSFET Switching

Electromagnetic emissions (EME) in power electronic converters primarily originate from the rapid voltage and current transitions that occur during MOSFET switching events [7]. Modern WBG devices, such as SiC MOSFETs, exhibit very high switching speeds, which may lead to excessive excitation of parasitic circuit elements and consequently increase EME if not properly controlled.

During turn-on and turn-off events, steep drain–source voltage ($\mathrm{d}v_{\mathrm{DS}}/\mathrm{d}t$) and drain current ($\mathrm{d}i_{\mathrm{D}}/\mathrm{d}t$) transitions excite parasitic elements within the device package, PCB layout, and commutation loop. Parasitic inductances and capacitances form resonant networks that give rise to voltage overshoot, ringing, and oscillations, which constitute sources of both conducted and radiated electromagnetic emissions [2].

- **Conducted EME -** Mainly associated with high-frequency current components flowing through the power and gate loops, which couple into the DC-link and external supply via parasitic impedances [7]. Conducted EME can be further classified into differential-mode (DM) and common-mode (CM) noise, each governed by distinct physical mechanisms associated with the switching transition, namely $\mathrm{d}i/\mathrm{d}t$ and $\mathrm{d}v/\mathrm{d}t$, respectively [22].
- **Radiated EME -** Arise when rapidly changing currents circulate in loops with non-negligible physical area, causing these loops to behave as unintended antennas that emit electromagnetic fields [22].

In addition to power-loop parasitics, gate-loop interactions also contribute to EME. Common-source inductance couples the drain current transient into the gate loop, potentially distorting the gate-source voltage [23]. Similarly, Miller capacitance (C_{GD}) injects displacement currents during voltage transitions. Furthermore, C_{GD} coupling can inject high-frequency disturbances into the gate during fast voltage transitions, further increasing the emission levels [17]. While this slows down the voltage and current commutation, it provides inherent damping that reduces parasitic ringing. Consequently, a specific amount of common source inductance is often deliberately designed into power modules to stabilize the switching behavior and enhance short-circuit robustness [23].

While faster switching may reduce switching losses, it simultaneously increases electromagnetic emissions, leading to a fundamental trade-off between efficiency and emission performance [3]. Effective emission reduction therefore requires coordinated optimization of gate drive parameters, device packaging, and layout design.

2.3.1 EME Figure of Merit for Power MOSFETs

Electromagnetic emissions (EME) can be broadly classified into radiated emissions (> 30 MHz) and conducted emissions (150 kHz–30 MHz) as discussed, with the latter representing a primary concern in automotive applications under standards such as CISPR 25 [24]. This EME compliance is assessed using standardized measurement setups, such as a Line Impedance Stabilization Network (LISN) for conducted emissions [22], with regulatory limits specified in dBμV for conducted noise and dBμV/m for radiated noise.

These limits are defined over distinct frequency bands associated with sensitive radio services. For instance, the FM broadcast band (76–108 MHz) limit imposed by CISPR 25 Class 5 represents one of the most challenging compliance targets for automotive power electronics, with an exceptionally stringent Average limit of 18 dBμV and a Peak limit of 38 dBμV [24]. Although the standards do not explicitly constrain switching slew rates ($\mathrm{d}v/\mathrm{d}t$ and $\mathrm{d}i/\mathrm{d}t$), the high-frequency harmonics generated by these fast transients are the dominant contributors to emissions within these tightly regulated bands.

To quantify these EME without relying exclusively on complex spectral analysis, a specific figure of merit (FOM) is considered in this thesis [25].

The FOM employed in this work is based on the *time–frequency co-spread* ($\Pi_{t\omega}$) of a switching transition. The co-spread jointly characterizes the temporal dispersion (σ_t) and the frequential dispersion (σ_ω) of a signal by accounting for their combined effect. These dispersions are direct analogues of the weighted standard deviation in statistics and serve as quantitative measures of the signal spread in the respective domains.

According to the Heisenberg-Gabor uncertainty principle, the product of the temporal and frequential dispersions is subject to a fundamental lower bound [25]:

$$\Pi_{t\omega} = \sigma_t \cdot \sigma_\omega \geq \frac{1}{2}. \tag{2.8}$$

This inequality implies that a physical limit exists on how strongly a signal (e.g., v_{DS}) can be localized in both the time and frequency domains simultaneously. The theoretical optimum, corresponding to the lower bound of $1/2$, represents the best possible trade-off between temporal and frequential spreading. This limit is theoretically achievable only when the switching transition waveforms, namely v_{DS} and i_{D}, follow the rising and falling edges of a Gaussian function during the turn-on and turn-off transitions [25].

2.3.2 Temporal and Frequential Dispersion

2.3.2.1 Temporal Dispersion

The temporal dispersion σ_t, defined as the standard deviation of the signal's distribution in the time domain, quantifies how the waveform is spread over time [25]. In this thesis, the HS and LS drain-source voltages (v_{DS}) are considered for its evaluation (Fig. 2.4). The mathematical formulation and numerical evaluation of this standard deviation are described in detail in Section 3.1.3.2. A larger value of σ_t indicates that the signal extends over a longer duration or exhibits a wider temporal spread.

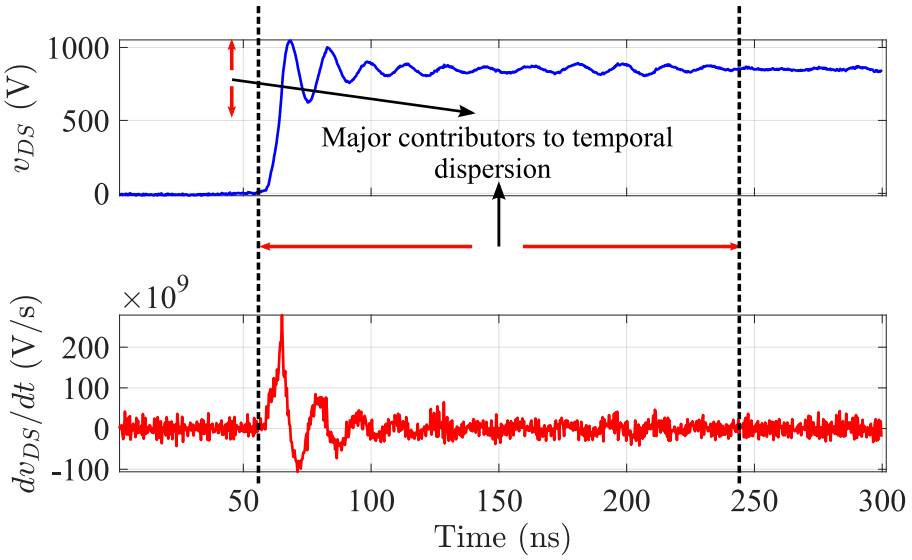


Figure 2.4: Temporal dispersion of a HS v_{DS} waveform during turn-on transition.

2.3.2.2 Frequential Dispersion

The frequential dispersion σ_ω characterizes the spread of the signal energy in the frequency domain and can be interpreted as a weighted standard deviation of the spectral energy distribution [25] (Fig. 2.5). The mathematical formulation and numerical evaluation of the frequential dispersion are described in detail in Section 3.1.3.2. A larger value of σ_ω indicates increased high-frequency content and a wider spectral spread.

In the frequency spectrum of the v_{DS}, two cutoff frequencies can be distinguished, which partition the spectrum into regions governed by different physical mechanisms of the switching process.

- **First cutoff frequency f_{cut1}:** The first cutoff frequency is associated with the periodicity of the switching signal and is primarily determined by the switching period T_s and the duty cycle D of the pulse-width modulated waveform. It is given by [26]

$$f_{\mathrm{cut1}} = \frac{1}{\pi T_s}. \tag{2.9}$$

  For frequencies below f_{cut1}, the spectrum is dominated by low-frequency components arising from the periodic nature of the switching waveform, with the harmonic amplitudes shaped by the duty cycle and the overall voltage swing.

- **Second cutoff frequency f_{cut2}:** The second cutoff frequency is governed by the effective transition time of the switching event, characterized by either the 10–90% rise time or the 90–10% fall time of the drain–source voltage (or equivalently, the drain current). It is defined as [26]

$$f_{\mathrm{cut2}} = \frac{1}{\pi \cdot 1.25 \cdot t_{10\text{–}90\%}}. \tag{2.10}$$

In the frequency range between f_{cut1} and f_{cut2}, the spectrum approximately follows a $-20\,\mathrm{dB/dec}$ slope, corresponding to the first-order behavior of the switching transition. Above

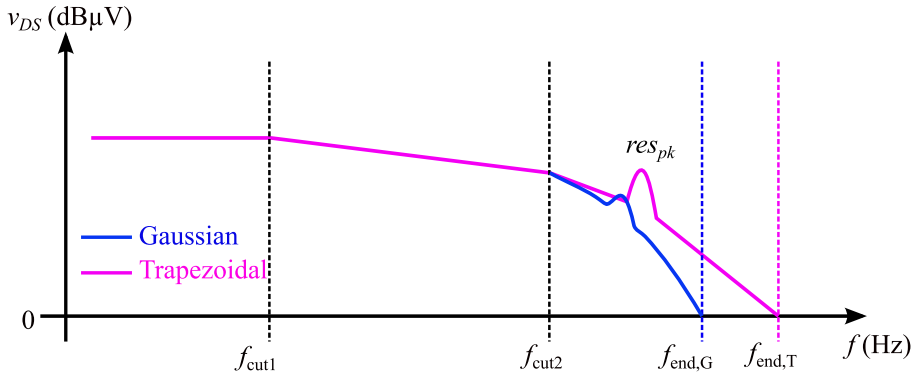


Figure 2.5: Spectral envelope of HS v_{DS} waveform during the turn-on transition [26].

f_{cut2}, the spectral decay becomes even more steeper, reflecting the smoothing effect of the finite voltage slope.

When the switching period (PWM frequency) is fixed, the second cutoff frequency f_{cut2} becomes the primary controllable parameter influencing the emission spectrum. Faster switching transitions, characterized by a smaller $t_{10–90\%}$, shift f_{cut2} toward higher frequencies, thereby increasing high-frequency spectral content and associated electromagnetic emissions.

2.3.3 Methods to Reduce EME

2.3.3.1 Reducing Temporal Dispersion

- **Shortening the effective transition time of the signal:** Reducing the $t_{10–90\%}$ decreases the temporal extent of the switching waveform and therefore lowers σ_t.
- **Suppressing post-transition ringing and overshoots:** Parasitic-induced ringing and overshoots extend the duration over which significant signal energy is present, redistributing energy beyond the nominal transition interval and increasing temporal dispersion. Effective damping of post-transition ringing and overshoots is therefore essential to limit the temporal spread of the waveform and minimize σ_t.

2.3.3.2 Reducing Frequency Dispersion

- **Shifting the cutoff frequency f_{cut2} to lower values:** Slower voltage transitions increase the effective transition time and therefore shift f_{cut2} toward lower frequencies. As a result, the high-frequency spectral content is reduced, leading to a narrower effective emission bandwidth while simultaneously mitigating voltage ringing and suppressing parasitic resonance peak ($V_{res,pk}$) in the frequency domain (Fig. 2.5). This approach corresponds to conventional EME mitigation strategies based on reducing dv/dt, at the expense of increased switching losses.
- **Increasing the spectral decay rate through Gaussian-shaped transitions:** Beyond the location of f_{cut2}, the *shape of the switching transition* becomes the dominant factor governing the spectral decay rate and, consequently, the overall frequential dispersion. For conventional trapezoidal switching waveforms, which are only first-order continuous,

the spectral envelope exhibits an asymptotic decay proportional to $1/f^2$ (approximately $-40\,\mathrm{dB/dec}$) until it reaches the noise floor or complete decay at $f_{\mathrm{end,T}}$ [26] (Fig. 2.5).

In contrast, Gaussian-shaped switching transitions, which are infinitely differentiable, exhibit a significantly steeper and curved decay compared to ideal or traditional trapezoidal waveforms [11] (Fig. 2.5). This rapid decay rate enables substantial suppression of high-frequency components for the same effective switching time, thereby reaching the noise floor faster at $f_{\mathrm{end,G}}$ and reducing the overall frequency dispersion (Fig. 2.5).

3 State-Space Simulation Framework of Power MOSFET

3.1 Simulation Framework for the SiC MOSFET

The state-space framework employed in this thesis is a physics-based simulation approach that predicts the transient behavior of power electronic switching cells by explicitly tracking the system's energy storage [27]. Central to this method are the state variables, which are defined as the specific quantities that store energy within the circuit's energy storage elements, namely:

- Voltages across capacitors ($u_{C,x}$).
- Currents flowing through inductors ($i_{L,x}$).

Instead of treating the circuit as a static netlist, the model determines the rate of change of these state variables as an explicit function of the current system state and external sources, allowing the evolution of the system quantities to be calculated directly via integration over time.

While SPICE relies on implicit, iterative numerical solving (such as Newton-Raphson) which can be time-consuming and prone to convergence issues when handling non-linear device capacitances, this state-space model uses an explicit formulation that avoids iterative loops. By directly calculating the time derivatives of all energy-storing elements and advancing the solution using standard numerical integration, the framework enables improved computational speed and numerical stability [27].

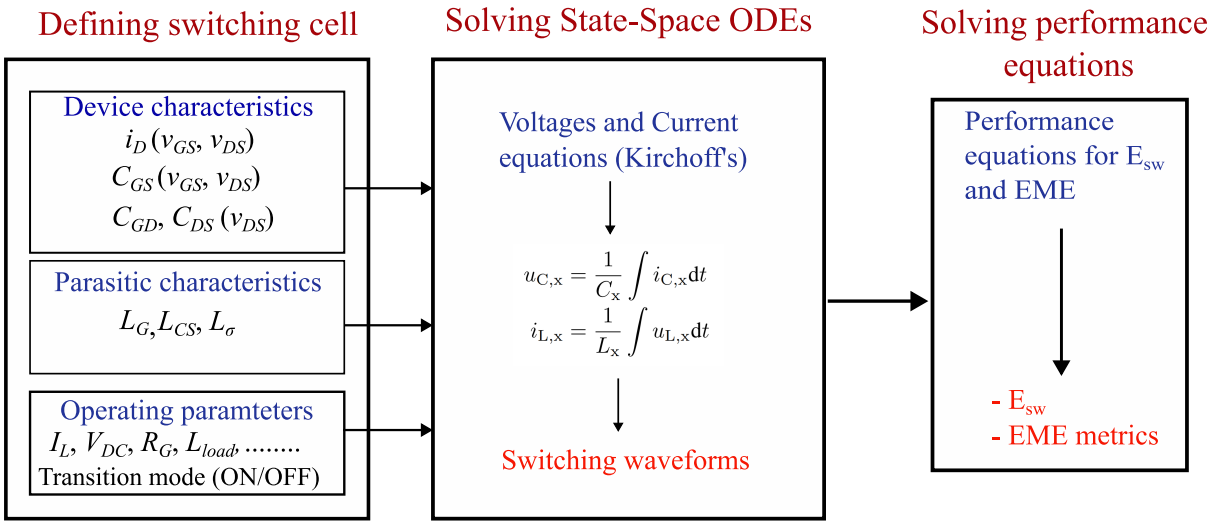


Figure 3.1: State Space Framework.

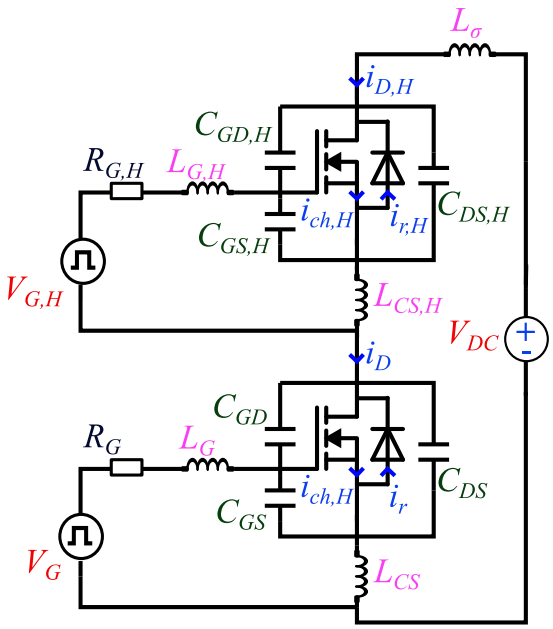


Figure 3.2: Simplified Switching cell Model [27].

This framework can be implemented in computational environments, such as MATLAB in this thesis, and enables efficient analysis and optimization of different combinations of semiconductor devices, package technologies, and gate-driving strategies.

The simulation framework is structured into three distinct stages (Fig. 3.1):

- Definition of the switching cell
- Numerical solution of the state-space ordinary differential equations (ODEs)
- Evaluation of performance-related quantities

3.1.1 Definition of the Switching Cell

The simulation framework begins with the parameterization of the switching cell, which consists of a low-side (LS) power MOSFET acting as the device under test (DUT), a high-side (HS) passive device, a DC-link voltage source, and gate-voltage sources, as shown in Fig. 3.2. Moreover, the switching cell model also includes the following elements:

- Nonlinear, voltage-dependent device capacitances.
- Device currents relevant to the switching transition.
- Parasitic inductances, including the gate-loop inductance (L_G), commutation-loop inductance (L_σ), and common-source inductance (L_{CS}).
- Gate resistances associated with the driving network.

Quantities associated with the high-side device are denoted by the subscript H (Fig. 3.2).

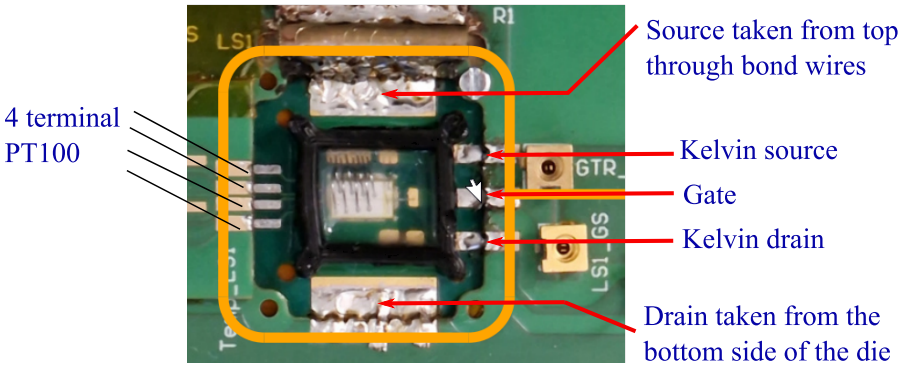


Figure 3.3: Bosch bare-die SiC MOSFET (BT2M1200009BOA) [29].

3.1.1.1 Power device

The devices used for State Space Simulation and experiments are Bosch SiC MOSFETs - BT2M1200009BOA [28] (Fig. 3.3). This MOSFET is a bare-die 1200 V SiC MOSFET based on Bosch's dual-channel trench gate technology [28], featuring bottom-side drain and top-side source connections realized via bond wires. A dedicated Kelvin source is provided to decouple the gate-drive loop from the high-current source path, reducing common-source inductance and enabling faster switching.

3.1.1.2 Gate-source Capacitance (C_{GS}) model

The dependence of the C_{GS} on both the instantaneous gate-source and drain-source voltages is accounted for in the implemented model,

$$C_{GS} = f(v_{GS}, v_{DS}). \tag{3.1}$$

To identify this voltage dependence, a combination of small-signal capacitance measurements and dynamic switching experiments using the double-pulse test (DPT) is employed, followed by parameter fitting.

Small-signal capacitance measurements (Device OFF):

Capacitance measurements are performed using a curve tracer (Keysight B1505A) with a small-signal AC excitation of 30 mV at 100 kHz. The following measurement configurations are used:

- C_{iss} **vs.** V_{DS}**:** V_{DS} swept from 0.1 to 1200 V (log scale) for $V_{GS} = 2, 1, 0, \ldots, -10$ V
- C_{GS} **vs.** V_{GS}**:** V_{GS} upsweep from −10 to +20 V with $V_{DS} \approx 0$, $I_D \approx 0$, and $I_G \approx 0$.
- C_{GD} **vs.** V_{GS}**:** V_{GS} upsweep from −10 to +20 V under identical quasi-static conditions.

Dynamic capacitance extraction using DPT (Device ON:)

The double pulse test (DPT) was performed at the following operating point: external turn-on gate resistance of $R_{\text{G,on}} = 220\,\Omega$, load current of $I_L = 180$ A, and DC-link voltage (V_{DC}) swept from 100 V to 850 V in 50 V increments. The evolution of the v_{GS} up to 18 V is inherently captured by the gate-driver dynamics.

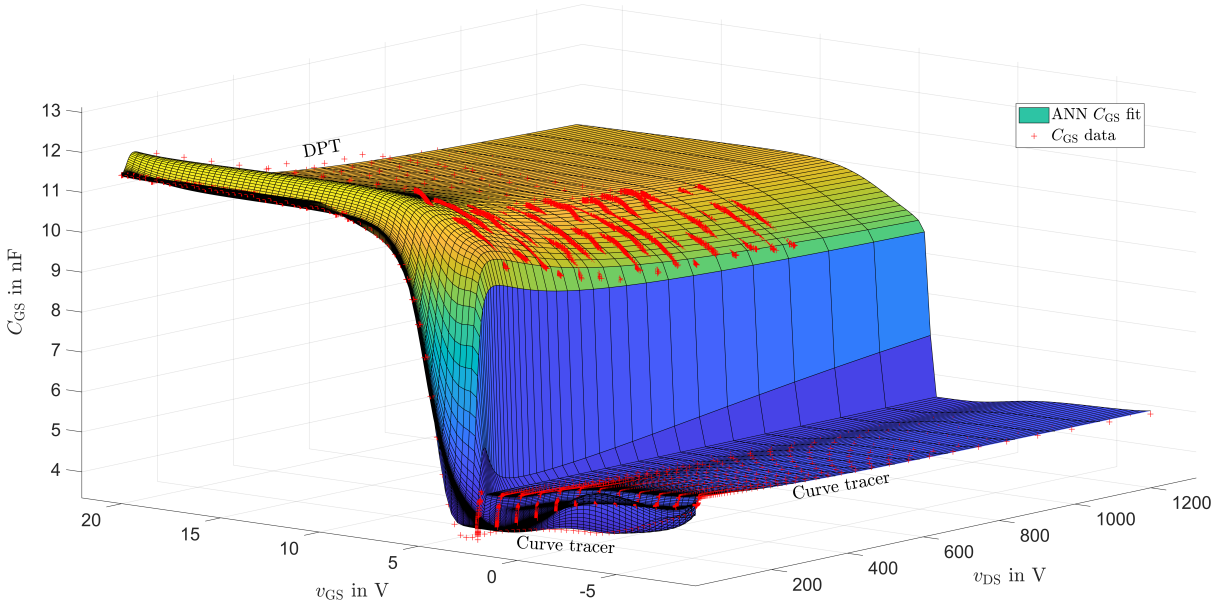


Figure 3.4: ANN-based fit of C_{GS} as a function of v_{GS} and v_{DS}.

During the ON transition, the gate current can be expressed as

$$i_G(t) = C_{GS}(v_{DS}) \frac{\mathrm{d}v_{GS}}{\mathrm{d}t} + C_{GD}(v_{DS}) \frac{\mathrm{d}v_{GD}}{\mathrm{d}t}. \tag{3.2}$$

At high v_{DS}, the input capacitance C_{iss} is dominated by C_{GS}, while C_{GD} remains comparatively small. As v_{DS} decreases during the turn-on transition, C_{GD} increases. While the specific slope of the Miller plateau is primarily governed by finite transconductance and the DIBL effect [15], the dominant capacitive feedback current is determined by C_{GD} [30]. Exploiting this separation of influence, C_{GS} is identified as a function of both v_{GS} and v_{DS} using experimental DPT data.

The dependency on v_{GS} is included to capture the non-linear variation of the input capacitance during the channel accumulation and inversion phases [15]. The extracted $C_{GS}(V_{GS}, V_{DS})$ data are subsequently used for parameter fitting of the capacitance model.

Artificial Neural Network (ANN)-Based Fit:

To evaluate C_{GS} for arbitrary combinations of v_{GS} and v_{DS}, a curve-fitting procedure is required for interpolation. Here, an ANN-based regression model is employed for fitting C_{GS}, resulting in a continuous nonlinear $C_{GS}(v_{GS}, v_{DS})$ model (Fig. 3.4) [12].

The measured v_{GS}, v_{DS}, and C_{GS} data are first normalized (e.g., to the [0,1] range) to improve numerical stability and training efficiency. A feedforward neural network is then implemented, with v_{GS} and v_{DS} as inputs and C_{GS} as the output. The network consists of multiple hidden layers employing sigmoid activation functions to capture strong nonlinear dependencies [12]. The ANN is trained iteratively on the preprocessed data by optimizing the internal weights and biases. Robust fitting is achieved using bisquare weighting, which ensures stable convergence of the ANN training process while reducing the influence of measurement outliers.

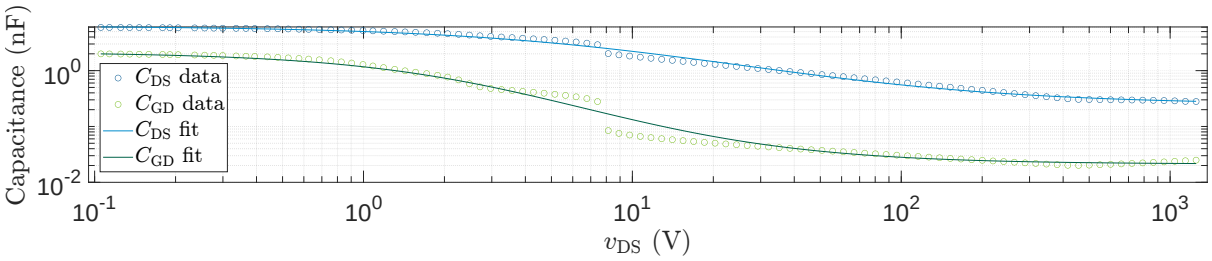


Figure 3.5: Tangent-based fit of C_{DS} and C_{GD} as a function of v_{DS}.

3.1.1.3 Drain–Source and Miller Capacitance Model (C_{DS} and C_{GD})

The dependence of the C_{GD} and C_{DS} on the instantaneous v_{DS} is accounted for in the model.

$$C_{GD}, C_{DS} = f(v_{DS}) \tag{3.3}$$

Similar to the C_{GS} characterization, small-signal capacitance measurements are performed using a curve tracer. The input capacitance C_{iss} and reverse transfer capacitance C_{rss} are measured as functions of V_{DS} by sweeping V_{DS} from 0.1 to 1200 V in logarithmic steps for $V_{GS} = 0, -1, \ldots, -10$ V.

Analytical fit:

C_{GD} and C_{DS} are modeled as strongly nonlinear functions of v_{DS}. A custom analytical formulation based on a hyperbolic tangent (tanh) function is employed to capture the pronounced voltage dependence observed in the measurements (Fig. 3.5) [31]. The model parameters are identified using nonlinear least-squares fitting, ensuring an accurate match between the analytical expressions and the experimentally measured capacitance data.

3.1.1.4 Drain current model

The drain current i_D is incorporated using a nonlinear model as a function of v_{GS} and v_{DS}.

$$i_D = f(v_{GS}, v_{DS}) \tag{3.4}$$

The drain current model is derived from static and dynamic measurements using a curve tracer and DPT measurements.

Static measurements:

- $R_{\mathrm{DS,on}}$ **extraction:** Output characteristics measured over $I_D = 0$ to 180 A in discrete steps for $V_{GS} = 13.5$ to 20 V (considering linear region).
- **Off-state breakdown:** V_{DS} is increased under off-state conditions while I_D is limited to 2 mA after the onset of breakdown.
- **Third-quadrant conduction:** Reverse-conduction characteristics obtained with $V_{GS} = -5\,V$ by sweeping V_{DS} in the reverse direction until $I_D = -180$ A.
- **Threshold voltage extraction:** Transfer characteristics measured by sweeping V_{GS} with $V_{DS} = V_{GS}$; V_{th} defined at $I_D = 25$ mA.

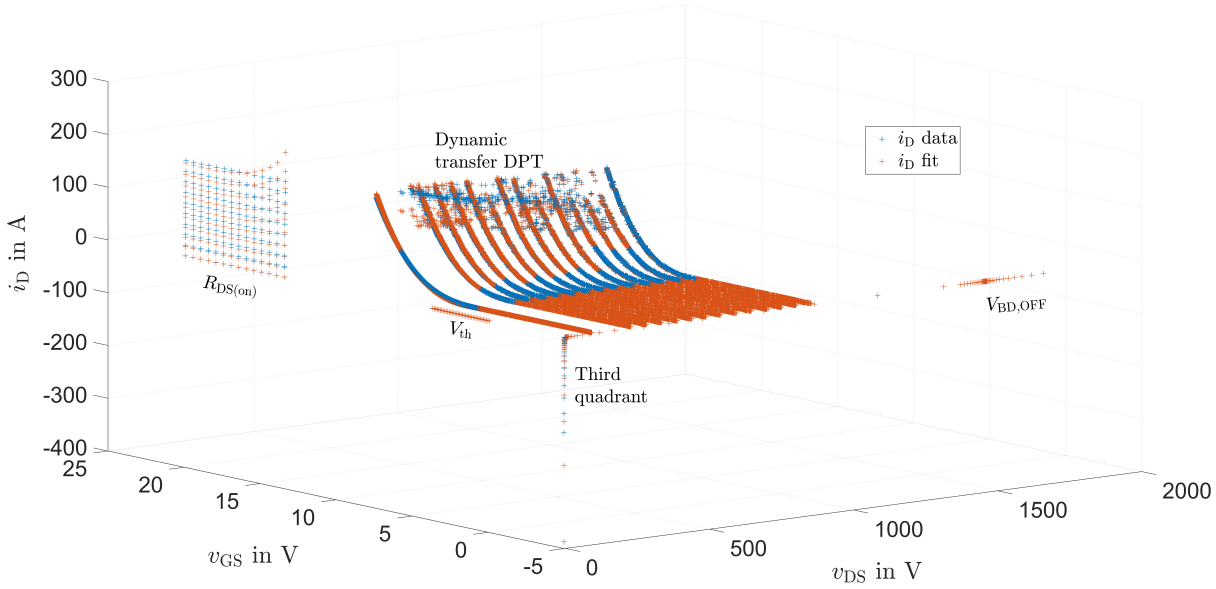


Figure 3.6: Analytical fit of drain current i_D as a function of v_{GS} and v_{DS}.

Dynamic transfer measurements:

Dynamic transfer characteristics are obtained from DPT measurements under the same operating conditions used for C_{GS} extraction, with $R_{\text{Gon}} = 220\,\Omega$, $R_{\text{Goff}} = 39\,\Omega$, $I_L = 180\,\text{A}$, and $V_{DC} = 100, 150, 200, \ldots, 850\,\text{V}$, where the V_{GS} evolution up to 18 V is inherently captured by the gate-driver dynamics.

Analytical fit:

An analytical drain-current model is derived using nonlinear curve fitting based on experimentally measured data (Fig. 3.6). The drain current i_D is represented by a nonlinear parametric formulation, with separate expressions for the relevant conduction regions [31]:

$$i_{\text{D}} = \begin{cases} \max\left(\min\left(i_{\text{ch,lin}}, i_{\text{ch,sat}}\right), 0\right), & \text{for } v_{\text{DS}} \geq 0 \\ \min\left(i_{\text{ch,lin}}, i_{\text{r}}\right), & \text{for } v_{\text{DS}} < 0 \end{cases} \tag{3.5}$$

with

First quadrant (forward conduction)[31]:

$$i_{\text{ch,lin}} = p_1\left(v_{\text{GS}} - p_3\right) \cdot v_{\text{DS}} \tag{3.6}$$

$$i_{\text{ch,sat}} = p_2\left(v_{\text{GS}} - p_3 \cdot (1 - p_4 v_{\text{DS}})\right)^M \cdot (1 + p_5 \cdot v_{\text{DS}}) \tag{3.7}$$

Third quadrant (reverse conduction) [31]:

$$i_{\text{r}} = p_6 \cdot \exp\left(-\frac{v_{\text{DS}} - p_7}{p_8}\right) \tag{3.8}$$

where $i_{\text{ch,lin}}$, $i_{\text{ch,sat}}$, and i_{r} denote the channel current in the linear region, the channel current in saturation, and the reverse-conduction (third-quadrant) diode current, respectively.

The parameters p_1–p_8 and M characterize the MOSFET behavior across its various operating regions [31]. Certain parameters, such as the threshold voltage p_3, are known and are therefore fixed during the fitting process. The remaining parameters are identified from the experimental data using nonlinear curve fitting, so that weighted least-squares error between the measured drain current and the modeled current $i_{D,\mathrm{fit}}$ is minimized.

3.1.2 Numerical Solution of the Coupled Switching Equations

Once the switching cell definition is completed, its dynamic behavior can be described using a generic state-space formulation. The fundamental structure of this state-space model is based on the energy-storing elements of the switching cell, namely all inductances and capacitances. Accordingly, the state variables are defined as the voltages across the linear and nonlinear capacitors, $u_{C,x}$, and the currents flowing through the parasitic and circuit inductances, $i_{L,x}$ [27]. These states are given by

$$u_{C,x} = \frac{1}{C_x} \int i_{C,x} \, \mathrm{d}t, \tag{3.9}$$

$$i_{L,x} = \frac{1}{L_x} \int u_{L,x} \, \mathrm{d}t. \tag{3.10}$$

The nonlinear device characteristics are embedded directly into the state-space model. Since these quantities depend only on the current state variables, the device models can be evaluated explicitly at every time step and naturally integrated into the state-space formulation [27]. To complete the state-space model, the inductance voltages $u_{L,x}$ and capacitance currents $i_{C,x}$ must be determined as functions of the state variables. This is achieved by applying Kirchhoff's voltage law (KVL) and Kirchhoff's current law (KCL) to the switching cell. The terminal voltages of the power device are constrained by [27]

$$u_m + u_{GS} - u_{DS} = 0, \tag{3.11}$$

while the currents at the switching node must satisfy [27]

$$i_G + i_D - i_{CS} = 0. \tag{3.12}$$

Consequently, the calculation of the inductance voltages and capacitance currents is performed in a manner that strictly preserves these constraints. In this way, a consistent and numerically stable state-space formulation of the switching cell is established [27]. The resulting set of nonlinear, time-varying equations is then evaluated using explicit numerical time integration with sufficiently fine time resolution to accurately capture fast switching transients [27]. At each time step, the state derivatives are evaluated from the current state, and the solution is advanced in time, ensuring a self-consistent evolution of all electrical quantities.

Through this numerical approach, all switching waveforms, including v_{DS}, i_D, v_{GS}, i_G, as well as the corresponding high-side voltages and currents, emerge naturally from the solution of the governing equations.

For example, v_{DS} can be obtained from the commutation loop dynamics described by [27]

$$V_{DC} - v_{DS} - L_\sigma \frac{\mathrm{d}i_D}{\mathrm{d}t} = 0, \tag{3.13}$$

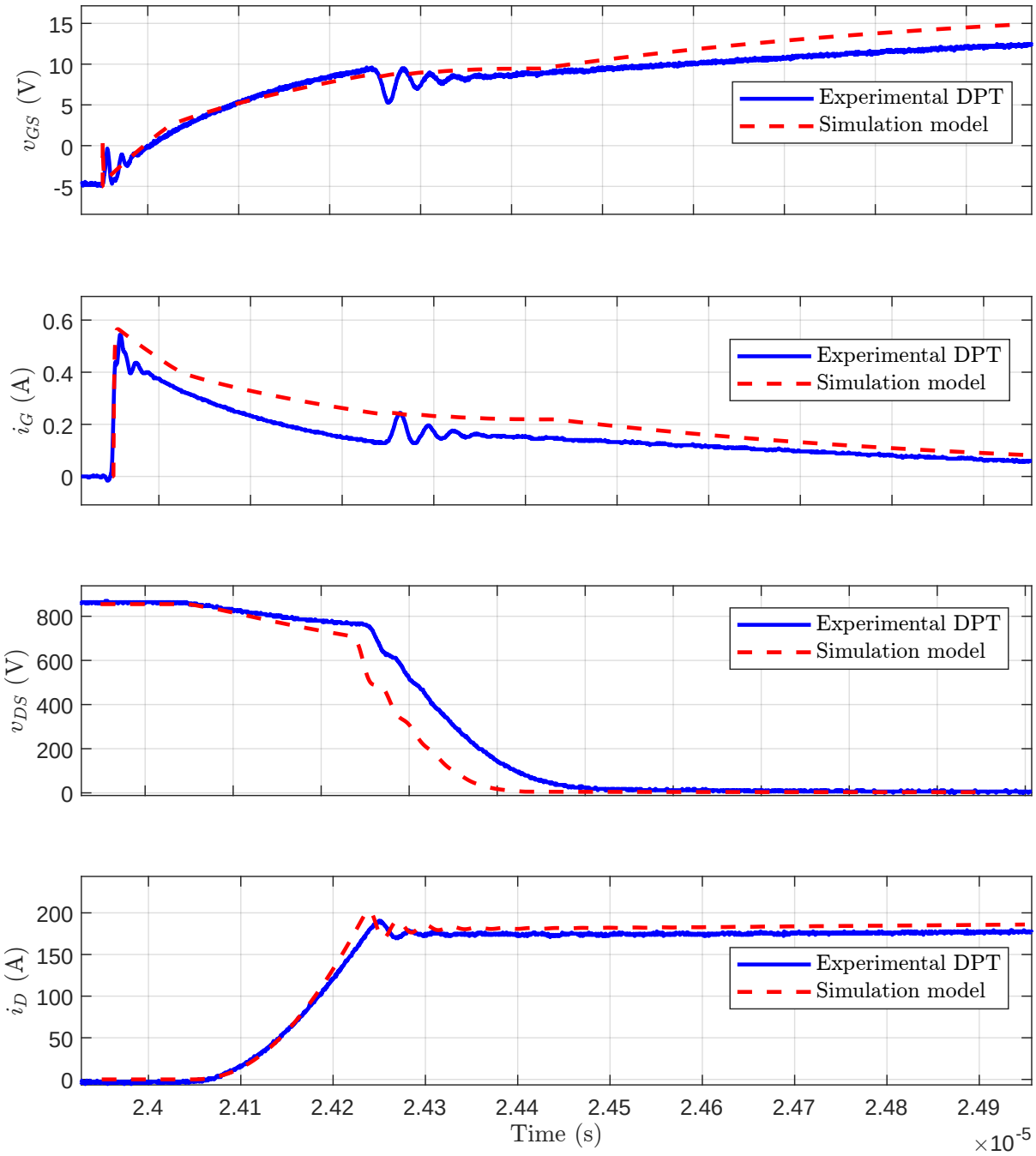


Figure 3.7: Comparison of switching waveforms between simulation and experimental DPT.

As a result, these simulated waveforms inherently capture the influence of parasitic inductances, nonlinear capacitances, and operating conditions, providing a physically consistent basis for subsequent switching loss and electromagnetic emission (EME) analysis.

3.1.2.1 Comparison of the Simulation Framework with the Actual Double-Pulse Test

Figure 3.7 presents a comparison between the simulation framework and the experimental measurements obtained from the double-pulse test (DPT). The experimental data were acquired using the hardware setup detailed in Chapter 5, with a fixed gate resistance of $R_G = 36\,\Omega$. Distinct

deviations in the transition speeds and waveform characteristics provide insight into the effects of parasitic elements that are not fully captured in the simulation results.

- The simulation predicts a steeper v_{DS} fall time, primarily attributed to inaccuracies in the modeling of C_{GS}. The experimental behavior suggests that the actual C_{GS} is higher than the fitted model prediction, leading to an incorrect Miller ratio (C_{GS}/C_{GD}). This discrepancy directly impacts the voltage slope (dv_{DS}/dt) by altering the dynamic balance of the capacitive currents [30].
- The experimental v_{GS} waveform exhibits a distinct voltage dip at the onset of the Miller plateau, which is not seen in the simulation. This disturbance is caused by the common-source inductance (L_{CS}) coupling the high di_D/dt transient into the gate loop [17]. Although the hardware utilizes a Kelvin Source connection, the setup inherently includes a mutual inductance of approximately 250 pH between the Kelvin path and the power loop.
- Furthermore, the experimental gate current (i_G) is lower in magnitude and exhibits oscillatory peaks during the switching transition. While the PCB trace resistance is negligible (in the range of mΩ), the reduced peak current and the observed ringing indicate the presence of unmodeled gate loop inductance (L_G) [32]. The oscillations in the middle of the i_G waveform arise from the resonance between this parasitic L_G and the input capacitance, excited by the displacement current injected through C_{GD} during the steep v_{DS} fall [17].

Despite these minor discrepancies, the simulated switching transitions exhibit waveform shapes and transition behaviors that closely match the experimental results. Consequently, this simulation framework can serve as a reliable tool for the development and optimization of the gate-shaping strategies.

3.1.3 Performance Equations

3.1.3.1 Switching Energy Definition

The switching energy (E_{sw}) of a power MOSFET is generally defined as the sum of the turn-on and turn-off energy components. In this work, however, the analysis is restricted to the *turn-on switching event*. Furthermore, the switching energy is evaluated exclusively for the low-side (LS) MOSFET, which acts as the device under test in the DPT configuration. This definition is applied consistently in both the state-space simulations and the experimental measurements.

The turn-on switching energy can be defined as the time integral of the instantaneous power during the turn-on event (see (2.5)),

$$E_{\mathrm{SW,ON}} = \int_{t_{\mathrm{start,on}}}^{t_{\mathrm{end,on}}} v_{DS}(t)\, i_D(t)\, \mathrm{d}t.$$

- The start time $t_{\mathrm{start,on}}$ is define, in this work, as the instant at which v_{GS} reaches 10% of the applied gate-drive voltage $(0.1 \times 18\,\mathrm{V} = 1.8\,\mathrm{V})$ [33].
- The end time $t_{\mathrm{end,on}}$ is defined, in this thesis, as the instant at which v_{DS} LS falls to 3% of the DC-link voltage $(0.03 \times 850\,\mathrm{V} = 25.5\,\mathrm{V})$.

This definition ensures that the integration interval fully captures the voltage-current overlap responsible for the turn-on switching losses.

3.1.3.2 EME calculation definition

Temporal dispersion:

The temporal dispersion σ_t quantifies how the drain–source voltage signal (v_{DS}) is distributed in time, as discussed in Chapter 2 (Fig. 2.4). It can be calculated as per [25]

$$\sigma_t = \sqrt{\frac{\sum_i (t_i - t_b)^2 \left|\frac{\mathrm{d}v_{DS}}{\mathrm{d}t}(t_i)\right|_\lambda^2 \Delta t}{\sum_i \left|\frac{\mathrm{d}v_{DS}}{\mathrm{d}t}(t_i)\right|_\lambda^2 \Delta t}} \tag{3.14}$$

The computation of the temporal dispersion σ_t proceeds as follows.

- A finite time window containing the relevant switching event is selected from $v_{DS}(t)$ waveform , including the post-transition ringing. Discrete time samples $\{t_i\}$ and corresponding voltage samples $\{v_{DS}(t_i)\}$ are extracted.
- The time derivative $\mathrm{d}v_{DS}/\mathrm{d}t$ is computed numerically and λ-normalized such that its integral over time equals unity. This removes amplitude dependence and isolates the temporal distribution of the switching activity. Consequently, the metric inherently captures post-transition ringing.
- The temporal barycenter t_b is calculated as the weighted mean time of the normalized switching activity [25],

  $$t_b = \sum_i t_i \left|\frac{\mathrm{d}v_{DS}}{\mathrm{d}t}(t_i)\right|_\lambda^2 \Delta t, \tag{3.15}$$

  representing the effective center of the voltage transition in time.
- A weighted standard deviation in time is then evaluated. In classical statistics, the weighted standard deviation σ of a discrete variable x_i with weights w_i is defined as [34]

  $$\sigma = \sqrt{\frac{\sum_i w_i (x_i - \mu)^2}{\sum_i w_i}}, \tag{3.16}$$

  where μ denotes the weighted mean. By direct analogy, σ_t replaces x_i with the time samples t_i, the weighted mean μ with t_b, and the statistical weights w_i with $|\mathrm{d}v_{DS}/\mathrm{d}t|_\lambda^2$, as defined in (3.14).

Short and well-damped transitions result in small values of σ_t, whereas extended transitions or post-transition ringing redistribute switching activity over time and increase the temporal dispersion.

Frequential dispersion:

The frequential dispersion σ_ω quantifies how the drain–source voltage switching activity is distributed in the frequency domain, as discussed in Chapter 2 (Fig. 2.5). The frequential dispersion can be defined as [25]

$$\sigma_\omega = \sqrt{\frac{\sum_k (\omega_k - \omega_b)^2 \, |V_\lambda(\omega_k)|^2 \; \Delta\omega}{\sum_k |V_\lambda(\omega_k)|^2 \; \Delta\omega}}, \tag{3.17}$$

where $V_\lambda(\omega)$ denotes the Fourier transform of the λ-normalized time derivative of the drain–source voltage.

The computation of the frequential dispersion σ_ω proceeds as follows.

- The same finite time window used for the temporal dispersion analysis, containing the relevant switching event of $v_{DS}(t)$, is selected. From this window, the λ-normalized voltage-slope signal $\mathrm{d}v_{DS}/\mathrm{d}t|_\lambda$ is obtained and transformed into the frequency domain using a discrete Fourier transform, yielding spectral samples $\{V_\lambda(\omega_k)\}$.
- The frequential barycenter ω_b is calculated as the weighted mean frequency of the normalized spectrum [25],

  $$\omega_b = \sum_k \omega_k \, |V_\lambda(\omega_k)|^2 \; \Delta\omega, \tag{3.18}$$

  which represents the effective center of the spectral energy distribution.
- A weighted standard deviation in frequency is then evaluated, analogous to the temporal dispersion as per (3.17) .

Fast switching transitions with steep voltage slopes shift spectral energy toward higher frequencies and increase σ_ω, whereas smoother transitions or waveforms with higher-order continuity exhibit a faster spectral decay and yield smaller values of σ_ω.

EME FOM (HGI):

As discussed in Chapter 2, the EME FOM is based on the time–frequency co-spread and is defined as [25]

$$\Pi_{t\omega} \; = \sigma_t \cdot \sigma_\omega. \tag{3.19}$$

In this thesis, the metric $\Pi_{t\omega}$ is referred to as the Heisenberg Gabor Inequality (HGI). The HGI is evaluated for the HS and LS v_{DS} during the turn-on switching transition.

3.2 Results of Simulation Framework with Fixed Gate Resistance

The double-pulse test simulation is performed using the simulation framework discussed above at the operating points specified in the Table 3.1. The performance metrics (switching loss and EME) are compared for operation with a fixed gate resistance of 5.1 Ω and 7.5 Ω. The switching loss and EME metric (HGI) are calculated as per Section 3.1.3.

Figure 3.8 shows the *turn-on* switching waveforms obtained from the DPT simulation for fixed gate-resistance operation at R_G = 5.1 Ω and 7.5 Ω. As expected, the voltage and current transitions during active turn-on are slower for the 7.5 Ω case compared to 5.1 Ω, due to the reduced

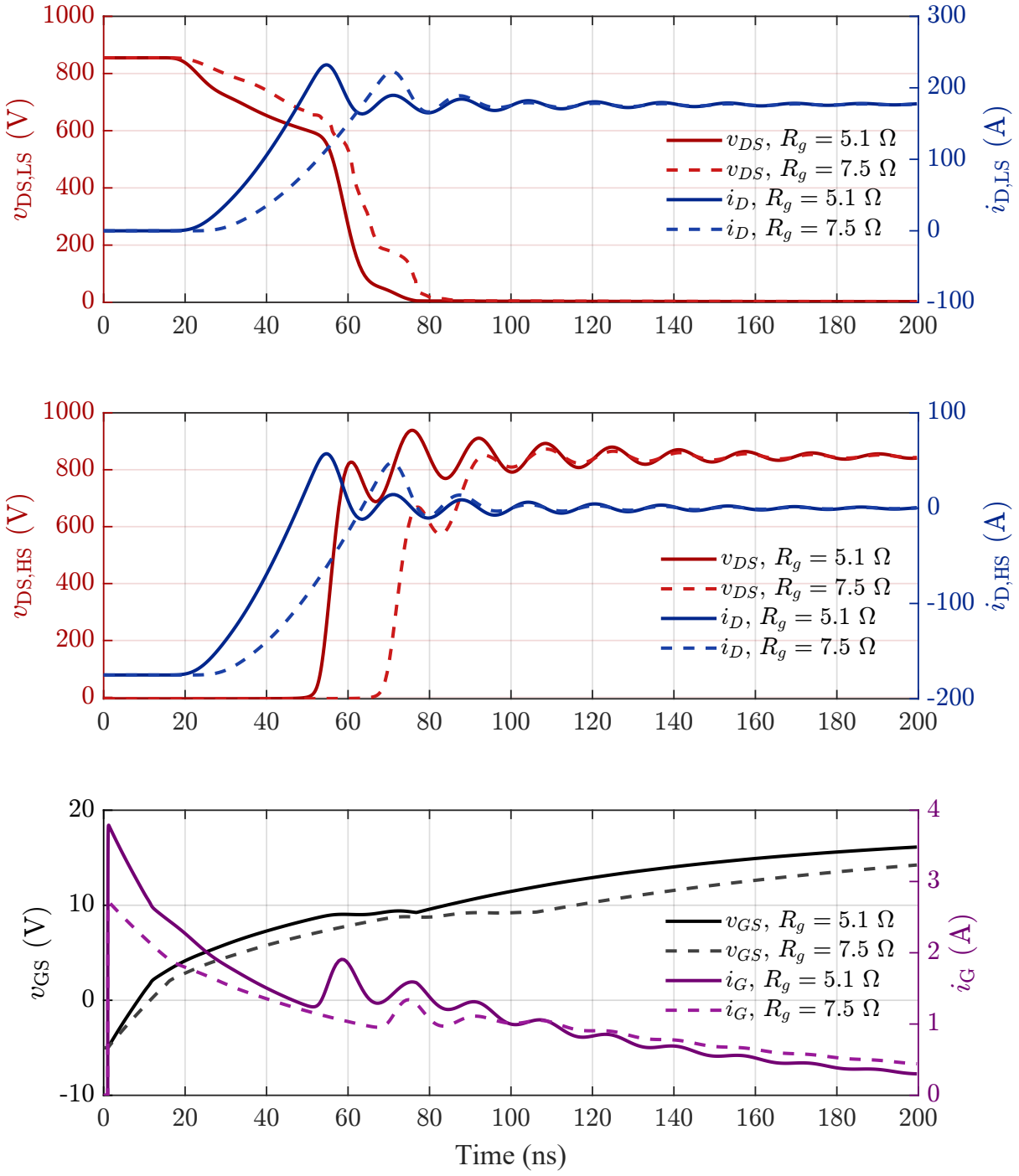


Figure 3.8: Switching waveforms for fixed R_G operation with $R_G = 5.1\ \Omega$ and $7.5\ \Omega$.

Table 3.1: Circuit Parameters and Operating Conditions

Symbol	**Description**	**Value**
Circuit Parameters:		
L_σ	Commutation loop inductance	20 nH
$R_{\mathrm{G,on}}$	External gate resistance (turn-on)	5.1 Ω / 7.5 Ω
$R_{\mathrm{G,off}}$	External gate resistance (turn-off)	15 Ω
$R_{\mathrm{G,int}}$	Internal gate resistance of device	0.931 Ω
Operating Conditions:		
V_{DC}	DC-link voltage	850 V
I_{Load}	Load current	175 A
L_{Load}	Load inductance	56 μH
$V_{\mathrm{GS,high}}$	Gate–source voltage (turn-on)	18 V
$V_{\mathrm{GS,low}}$	Gate–source voltage (turn-off)	−5 V

Table 3.2: Comparison of Switching Performance for Different Gate-Drive Methods at the operating conditions in Table 3.1

Method	E_{sw} (μJ)	t_{on} (ns)	$\mathrm{HGI_{LS}}$	$\mathrm{HGI_{HS}}$
Fixed R_G (5.1 Ω)	2805.9	61.41	2.16	6.21
Fixed R_G (7.5 Ω)	3870.7	80.73	2.03	3.49

initial gate charging current (i_{G}) caused by the higher gate resistance. The prolonged transition interval results in increased turn-on switching losses for the 7.5 Ω operation (Table (3.2)).

In contrast, the lower gate resistance of 5.1 Ω leads to a higher i_{G}, resulting in faster voltage and current transitions. This results in increased $\mathrm{d}v_{DS}/\mathrm{d}t$ and $\mathrm{d}i_D/\mathrm{d}t$, which lead to increased v_{DS} overshoot and ringing (Figure 3.8). In particular, the rapid change in current produces an additional voltage drop across the parasitic inductances (see (2.5)) while the fast voltage transition excites resonances with the parasitic capacitances. The combined effect of these parasitic inductance-capacitance interactions leads to voltage overshoot and ringing during the switching transient. These effects are reflected as increased HGI values in Table (3.2).

The gate current waveform exhibits a pronounced peak at the onset of active turn-on, corresponding to the rapid charging of the MOSFET gate input capacitance. This is followed by a gradual reduction in gate current as the device enters the Miller plateau. However, during the steep v_{DS} transition, a transient oscillation is observed in i_G (Figure 3.7). This phenomenon arises from the resonance between the gate loop inductance (L_G) and C_{iss}, which is excited by the displacement current injected through C_{GD} [17].

An important observation is that the gate current follows a natural RC-type decay after the initial peak, indicating that the fixed gate-resistance drive does not fully utilize the available gate-drive capability throughout the transition (Figure 3.8). Improved control of the gate current profile, such as through active gate current shaping, can potentially reduce voltage overshoot and ringing while maintaining low turn-on switching losses.

4 Gate Current Optimization Framework

In conventional fixed gate resistance driving of MOSFETs, the gate current (i_G) is initially high and passively decays over time due to the RC behavior of the gate loop, with its peak magnitude determined solely by the selected gate resistance [3] (Fig. 3.8). Once the gate resistance is fixed, no further shaping of the gate current waveform is possible during the switching transient. This limitation can be addressed by employing current source gate drivers, such as the EG120, which allow different gate current levels to be applied over predefined time intervals [10].

To determine optimized gate-current profiles that minimize switching loss (E_{sw}) and electromagnetic emissions (EME), which has to be applied to these current-source gate drivers, a genetic algorithm is employed in this thesis [12] (Fig. 4.1). In addition, the practical realization of these optimized profiles is investigated, and the associated implementation constraints are integrated directly into the optimization framework.

4.1 Cost Function

The genetic algorithm minimizes a normalized cost function f_{cost}, in which the weighting factors w_1 and w_2 define the relative emphasis between E_{sw} and EME minimization. The overall cost function is expressed as [12]

$$f_{\text{cost}} = \frac{w_1}{w_1 + w_2}\,\eta_{\text{SW}} + \frac{w_2}{w_1 + w_2}\,\eta_{\text{EME}} + p, \tag{4.1}$$

where η_{sw} and η_{EME} denote the normalized switching-loss and EME indices, respectively, and p is a penalty term accounting for V_{DS} overvoltage.

The normalized switching-loss index is defined as [12]

$$\eta_{\text{SW}} = \frac{E_{\text{sw}} - E_{\text{sw,min}}}{E_{\text{sw,max}} - E_{\text{sw,min}}} \tag{4.2}$$

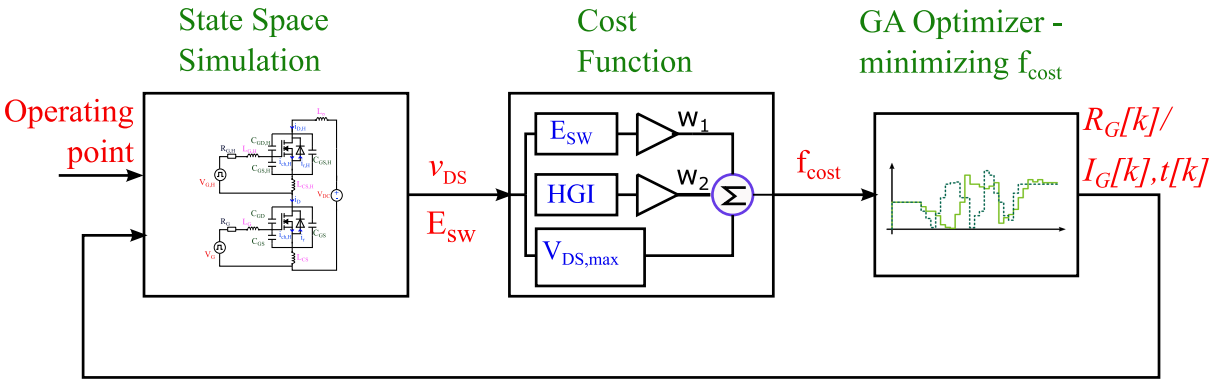


Figure 4.1: Gate Current Optimization Framework [12].

Similarly, the normalized EME index is given by [12]

$$\eta_{\mathrm{EME}} = \frac{(\mathrm{HGI_{LS}} - \mathrm{HGI_{LS,min}}) + (\mathrm{HGI_{HS}} - \mathrm{HGI_{HS,min}})}{(\mathrm{HGI_{LS,max}} - \mathrm{HGI_{LS,min}}) + (\mathrm{HGI_{HS,max}} - \mathrm{HGI_{HS,min}})}, \tag{4.3}$$

where $\mathrm{HGI_{LS}}$ and $\mathrm{HGI_{HS}}$ represent the EME figures of merits (FOMs) associated with the LS and HS switches, respectively.

The penalty term p is defined as [12]

$$p = \begin{cases} 0, & \text{if } V_{\mathrm{DS,max}} \leq 1200\ \mathrm{V}, \\ V_{\mathrm{DS,max}} - 1200\ \mathrm{V}, & \text{if } V_{\mathrm{DS,max}} > 1200\ \mathrm{V}. \end{cases} \tag{4.4}$$

This penalty term constrains the genetic algorithm against solutions that would otherwise violate the maximum allowable drain-source voltage overshoot exceeding the rated maximum voltage of 1200 V.

The weighting factors considered in this these are

$$[w_1,\ w_2] = \{[1,1],\ [9,1],\ [1,9]\}, \tag{4.5}$$

with the following interpretations:

- [1,1]: equal importance assigned to switching-loss minimization and EME reduction.
- [9,1]: increased emphasis on switching-loss reduction.
- [1,9]: increased emphasis on EME reduction.

4.2 EG120 Programmable Gate Driver

To enable real-world implementation of optimized gate-current profiles, the EG120 programmable gate driver is employed. Unlike conventional gate drivers that rely on a fixed gate resistance, the EG120 allows active shaping of the gate current during switching transitions. This is achieved by dynamically controlling the gate-driver output stage, enabling user-defined current profiles composed of multiple segments with independently adjustable amplitudes and durations [10].

The EG120 programmable gate current waveform consists of up to six discrete current amplitude levels (for ON-transition) applied over five independently configurable time intervals, allowing piecewise control of the switching transient (Fig. 4.2). The driver implements a single gate-drive channel with a total drive capability of 10 A, internally split into four synchronized outputs providing ±2.5 A each, enabling flexible and high-current gate control [10].

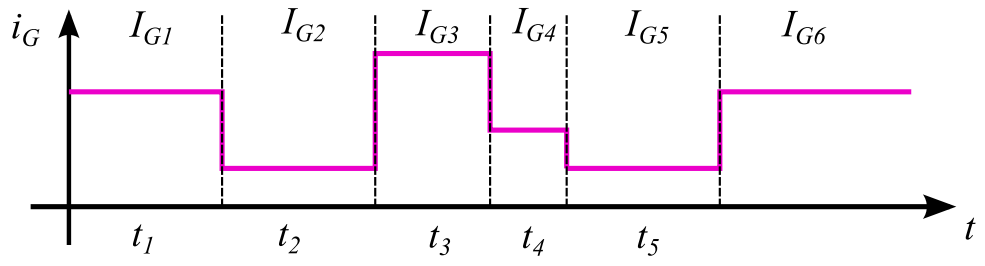


Figure 4.2: EG120 Gate current profile - ON transition.

4.3 Optimization of Gate-Current Profiles Using a Genetic Algorithm

4.3.1 Method 1: Continuous Gate-Resistance Profile Optimization

In the first approach, the genetic algorithm (GA) is employed to optimize an approximately continuous, time-varying gate-resistance profile $R_G[k]$ during the switching transient [12].

The gate-resistance profile is parameterized using $r_{\text{steps}} = 100$ independent resistance values uniformly distributed over a time window of $t_{\text{steps}} = 50$ ns starting at the switching instant. Although discretized, the high temporal resolution yields a dense parameterization that closely approximates a continuous resistance trajectory, thereby enabling smooth shaping of the gate current throughout the switching event. To ensure physically meaningful solutions and stable convergence of the GA, lower and upper bounds are imposed on the resistance values,

$$r_{\min} = 0.5\ \Omega \quad \text{and} \quad r_{\max} = 20\ \Omega.$$

In addition, an initial resistance value $r_{i,\text{ref}}$ within this range is used to initialize the GA population. This initialization accelerates the search toward practically relevant solutions.

Given the high dimensionality of the optimization problem, a large population size and multiple generations are employed to sufficiently explore the solution space [12]:

- Population size: 200 individuals
- Maximum number of generations: 50

E_{sw} is normalized using reference minimum and maximum values, $E_{\text{sw,min}}$ and $E_{\text{sw,max}}$, which are obtained from reference simulations using fixed gate-resistance values corresponding to $r_{\min}$ and $r_{\max}$, respectively (see (4.2)).

Similarly, EME contribution is quantified using the EME FOMs (HGI_{LS} and HGI_{HS}). The lower bounds of the HGI metric ($\text{HGI}_{\min}$) are fixed at 0.5, which arises from the fundamental time-frequency limitation imposed by the Heisenberg uncertainty principle (see (2.8)). The corresponding upper bounds ($\text{HGI}_{\max}$) are determined using Bayesian optimization by identifying the worst-case HGI values within the defined gate-resistance range [12] (see (4.3)).

Due to the lack of explicit hardware constraints on temporal resolution and gate-current slew rate, the optimized resistance trajectory will contain rapid or continuous variations that might not be realizable with practical gate drivers. Nevertheless, this method provides valuable insight into the theoretically optimal shaping of the gate current and establishes an upper performance bound for comparison with more practically constrained optimization approaches.

4.3.1.1 Results and Inference

The optimized gate-resistance profile simulation is performed at the same operating point and using identical circuit parameters as the corresponding fixed gate-resistance (R_G) case, thereby ensuring a fair and consistent comparison (Table 4.1). Fig. 4.3 shows the resulting waveforms for a weighting factor of [1,1], corresponding to equal emphasis on E_{sw} and EME. The switching losses, EME metrics, and drain-source voltage overshoot ($V_{DSHS,OS}$) obtained from the

Table 4.1: Circuit Parameters and Operating Conditions

Symbol	Description	Value
L_σ	Commutation loop inductance	20 nH
V_{DC}	DC-link voltage	850 V
I_{Load}	Load current	175 A
L_{Load}	Load inductance	56 μH
$v_{\mathrm{GS,high}}$	Gate–source voltage (turn-on)	18 V
$v_{\mathrm{GS,low}}$	Gate–source voltage (turn-off)	−5 V

Table 4.2: Comparison of Switching Performance for Different Gate-Drive Methods at the operating point in Table 4.1

Method	E_{sw} (μJ)	t_{on} (ns)	$\mathrm{HGI}_{\mathrm{LS}}$	$\mathrm{HGI}_{\mathrm{HS}}$	$V_{DSHS,OS}$ (V)
Fixed R_G (5.1 Ω)	2805.9	61.41	2.16	6.21	78
Fixed R_G (7.5 Ω)	3870.7	80.73	2.03	3.49	22
GA cont. [1,1] R_G profile	**2011.6**	44.86	**2.51**	**0.52**	**7**

fixed-R_G simulations are compared with those of the optimized gate-current–driven simulation (Table 4.2). All quantities are calculated in accordance with the formulations defined in Section 3.1.3.

Compared to fixed-R_G switching, the optimized transition is faster, resulting in reduced switching energy while simultaneously improving EME performance, particularly on the high-side (HS), which is the dominant contributor (Table 4.2). Despite the increased switching speed, no adverse effects such as voltage overshoot or ringing are observed (Fig. 4.3).

The HS v_{DS} exhibits an almost Gaussian-shaped transition without overshoot, while the LS i_D remains free of oscillations. This confirms that the voltage transition is not abrupt but instead features a gradual acceleration and deceleration, significantly reducing high-frequency spectral content and limiting excitation of the commutation loop resonance.

Gate-Current Waveform Shaping Mechanism:

Phase I: In the first phase, i_G follows the same pattern as in fixed gate-resistance operation until the device is turned on.

Phase II:

In this phase, i_G exhibits a bell-shaped pattern. The increased gate current rapidly charges the gate capacitance, raising the gate potential and enabling the MOSFET to take over the load current. The resulting rise in $di_{D,\mathrm{LS}}/dt$ induces a voltage drop across the commutation loop inductance (Fig. 4.3). Consequently, switching losses are lower during this current commutation interval due to this lower voltage drop. Furthermore, the subsequent voltage commutation phase is shortened, as the output capacitance (C_{oss}) of LS is already partially discharged.

Phase III: Following the initial acceleration, i_G is intentionally reduced and maintained at a low, nearly constant level. During this interval, $\mathrm{d}i_{D,\mathrm{LS}}/\mathrm{d}t$ decreases in a controlled manner, leading to a reduction of the inductive voltage drop (Fig. 4.3). This behavior corresponds to the flattening and slight increase observed in $v_{DS,\mathrm{LS}}$ waveform.

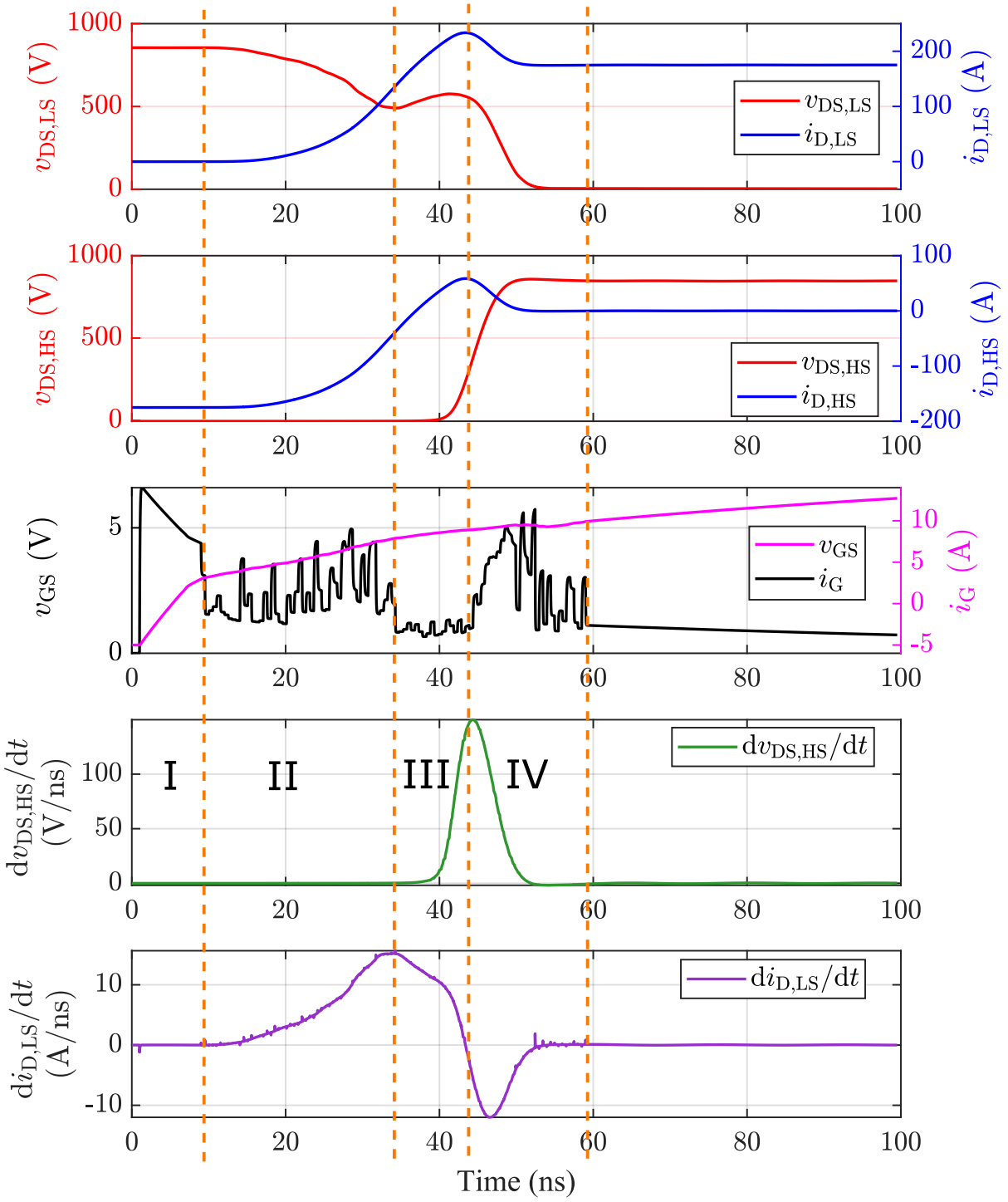


Figure 4.3: Switching waveforms for GA-optimized R_G profile with weight factors [1,1].

Phase IV: Once the current transition is complete, a second bell-shaped gate-current pulse is applied. This increase in i_G primarily drives the Miller plateau, enabling rapid v_{DS} commutation (Fig. 4.3). Since the current overlap has already been minimized in the preceding phases, the accelerated voltage transition shortens the overall switching event and further reduces switching energy. Importantly, the high-side v_{DS} transition remains free of overshoot and ringing, as parasitic dynamics are no longer strongly excited.

Overall, the optimized i_G shaping enables local acceleration and deceleration of the switching process, achieving faster switching with lower switching losses and improved EME performance compared to conventional fixed R_G gate-driving.

4.3.1.2 Discretising continuous gate current waveform

The optimized gate-current profile is nearly continuous and therefore cannot be directly implemented in hardware, as the EG120 gate driver supports only a finite set of discrete current levels and transition instants. (Fig. 4.2).

To enable practical implementation, the continuous optimized gate-current waveform can be discretized by segmenting it into a limited number of time intervals. Within each interval, a representative gate-current value is selected, typically using the mean value of the optimized waveform over that interval (Fig. 4.4). The segmentation is performed at physically meaningful locations ensuring that the dominant switching dynamics are preserved.

- **Event 1:** Onset of channel formation, defined by v_{GS} exceeding the threshold voltage.
- **Event 2:** Beginning of current commutation, indicated by i_D rising above 0.5% of the load current.
- **Event 3:** Instant of maximum $\mathrm{d}i_D/\mathrm{d}t$, after which the current slew rate decreases.
- **Event 4:** Completion of drain current commutation.
- **Event 5:** Completion of voltage commutation, defined by v_{DS} falling below 1% of its maximum value.

Some of these time intervals can be combined or extended, enabling an approximate reproduction of the optimized i_G waveform using a reduced number of discrete segments compatible with practical gate-driver constraints.

4.3.2 Method 2: Discrete Gate-Current Profile Optimization with EG120 Constraints

In the second approach, the genetic algorithm directly optimizes a discrete gate-current profile $I_G[k]$ that is explicitly constrained by the waveform generation capabilities of the EG120 programmable gate driver [10]. In contrast to Method 1, the gate current itself is treated as the optimization variable rather than being shaped indirectly through a gate-resistance profile.

The gate-current profile is parameterized using a finite number of constant current segments and associated time intervals. Specifically, the optimized waveform consists of four discrete current levels and three time intervals relative to the switching event.

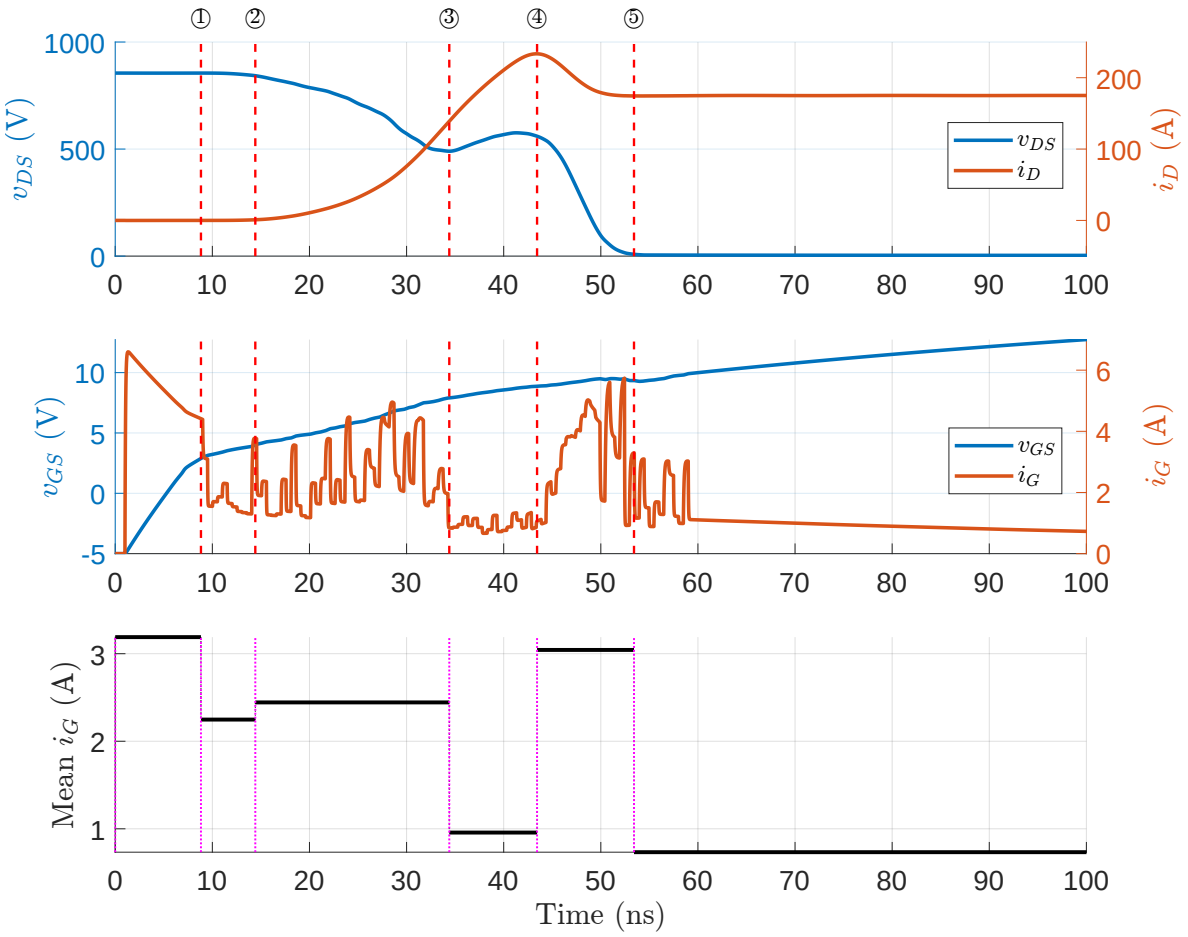


Figure 4.4: Event-based discretized optimized gate-current waveform.

$$I_G[k] = \{I_1, I_2, I_3, I_4\}, \quad T_k = \{T_1, T_2, T_3\}.$$

Each current level is applied over a corresponding time interval, forming a piecewise-constant gate-current waveform that directly resembles the programmable gate-current profiles supported by the EG120 (Fig. 4.2). To ensure physically realizable and hardware-compatible solutions, explicit bounds are imposed on both the gate-current amplitudes and the time intervals.

$$I_{G,\min} = 0.1 \text{ A}, \qquad I_{G,\max} = 5 \text{ A},$$

The discrete current segments are separated by a minimum time interval of $\Delta T_{\min} = 10\,\text{ns}$, adopted as a realistic lower bound to ensure practical implementation feasibility. To enhance convergence behavior, the initial population of the Genetic Algorithm (GA) is initialized using reference values for both the current levels and the transition times.

As in Method 1, the cost function consists of normalized switching energy and EME using the same formulation. A key feature of Method 2 is its substantially reduced computational complexity. Owing to the significantly lower number of optimization variables (only seven variables in total) compared to the high-dimensional resistance trajectory in Method 1, the GA converges considerably faster, allowing the population size to be reduced. As a result, the total optimization time is reduced from approximately 20 hours per simulation in Method 1 to about

Table 4.3: Comparison of t_{on} and E_{sw} for different simulation methods at the operating point in Table 4.1, referenced to $R_G = 5.1\,\Omega$

Method	t_{on} (ns)	E_{sw} (μJ)	ΔE_{sw} (%)
Fixed R_G (5.1 Ω)	61.41	2805.9	–
Fixed R_G (7.5 Ω)	80.73	3870.7	+38.0
GA cont. [1,1] R_G profile	44.86	**2011.6**	**−28.3**
GA EG120 [1,9] i_G profile	51.54	**2468.2**	**−12.0**
GA EG120 [9,1] i_G profile	38.98	**1396.7**	**−50.2**

Table 4.4: Comparison of HGI and $V_{DS\,HS,OS}$ for different simulation cases at the operating point in Table 4.1, referenced to $R_G = 5.1\,\Omega$.

Method	HGI_{LS}	HGI_{HS}	ΔHGI_{HS} (%)	$V_{DS\,HS,OS}$ (V)	$\Delta V_{DS\,HS,OS}$ (%)
Fixed R_G (5.1 Ω)	2.16	6.21	–	78	–
Fixed R_G (7.5 Ω)	2.03	3.49	−43.8	22	−71.8
GA cont. [1,1] R_G profile	2.51	**0.52**	**−91.6**	**7**	**−91.0**
EG120 [1,9] I_G profile	3.03	**0.53**	**−91.5**	**3**	**−96.2**
EG120 [9,1] I_G profile	9.99	3.06	−50.7	324	+315.4

6 hours in Method 2. This reduced runtime enables efficient exploration of multiple operating points and weighting combinations.

4.3.2.1 Results and Inference

This section presents the switching waveforms obtained using the EG120-based gate-driver optimization algorithm (Method 2). Figures 4.5 and 4.6 show the resulting waveforms for weighting factors of [1,9] and [9,1] respectively, as defined in Section 4.1.1. Table 4.5 presents the results of the genetic algorithm under EG120 constraints, including the optimized gate-current amplitudes and their corresponding timing values. The values shown correspond to the period after the MOSFET turn-on. Here, t_1 represents the duration for which I_{G1} is applied; similarly, t_2 and t_3 correspond to the application durations of I_{G2} and I_{G3}, respectively. The current level I_{G4} is applied for the remaining interval until the switching transition is complete, or until no further gate current is required.

The corresponding switching losses and electromagnetic emission (EME) metrics are evaluated and compared for both cases (Tables 4.3 and 4.4). All quantities are calculated in accordance with the formulations defined in Section 3.1.3.

The relative switching-loss change is calculated as

$$\Delta E_{sw,[9,1]} = \frac{E_{sw,[9,1]} - E_{sw,5.1}}{E_{sw,5.1}} \times 100 = \frac{1396.7 - 2805.9}{2805.9} \times 100 = -50.2\%. \tag{4.6}$$

Similar to (4.6), all relative switching-loss and EME performance metrics reported in this thesis are calculated with respect to the fixed gate-resistance configuration with $R_G = 5.1\,\Omega$.

Table 4.5: EG120 gate-current profiles for turn-on (Simulation)

Configuration	I_{G1} (A)	I_{G2} (A)	I_{G3} (A)	I_{G4} (A)	t_1 (ns)	t_2 (ns)	t_3 (ns)
GA EG120 [9,1] i_G profile	4.998	2.1917	4.835	4.835	10.0	18.26	9.01
GA EG120 [1,9] i_G profile	3.08	0.753	2.768	3.22	17.4	13.71	12.01

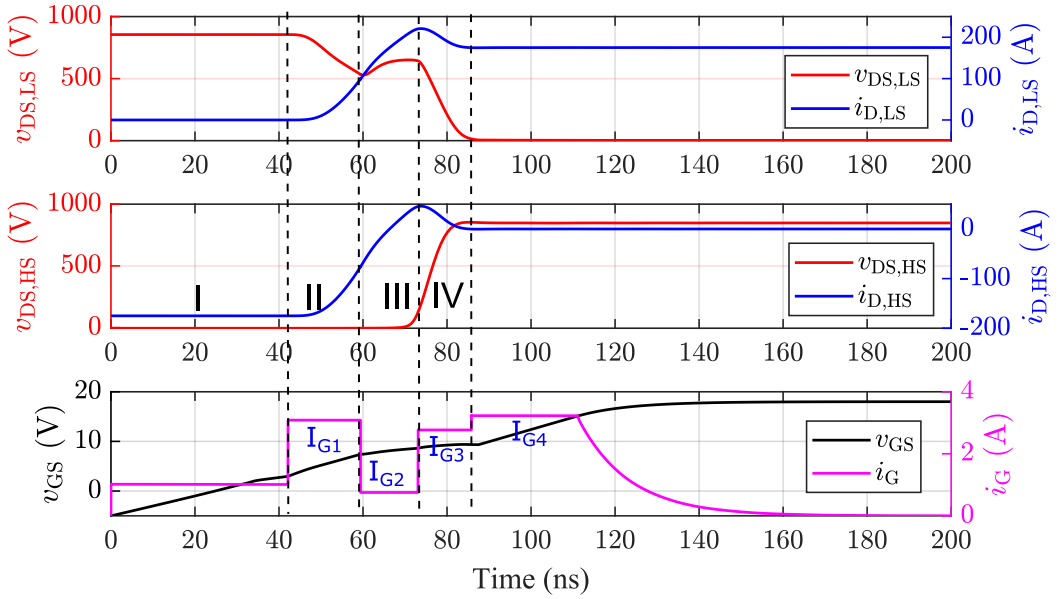


Figure 4.5: GA-optimized switching waveforms with EG120-constrained I_G profile - [1,9].

Case 1 - [1,9]:

The switching waveforms obtained in this case are similar to those obtained with a continuously optimized gate resistance profile, except for the i_G waveform, which is discrete and square-shaped rather than exhibiting a two bell-shaped profile (Fig. 4.5). The HS v_{DS} exhibits a nearly ideal Gaussian-shaped transition with no observable overshoot or ringing, as expected for a case optimized primarily for EME. This results in an HGI_{HS} value of 0.53 (nearly equal to the ideal value of 0.5), corresponding to a 91.5% reduction compared to the fixed $R_G = 5.1\,\Omega$ case (Table 4.4). The switching loss is also reduced by 12% compared to the fixed $R_G = 5.1\,\Omega$ case. However, it remains higher than that of the continuous optimization case due to the inclusion of EG120 constraints into the algorithm in this case (Table 4.3).

Case 2 - [9,1]:

This switching scenario achieves the lowest switching loss (1396.7 μJ) among all evaluated cases (Tables 4.3), representing a 50 2% reduction in switching loss compared to the fixed $R_G = 5.1\,\Omega$ case. However, its EME performance is degraded, as shown by the significant voltage overshoot and ringing. This behavior arises because the gate current I_G remains high throughout the switching transition, with a minimum value of approximately 2 A, resulting in very rapid commutation (Fig. 4.6). Furthermore, the genetic algorithm exploits the maximum allowable v_{DS} constraint of 1200 V, which leads to a pronounced voltage overshoot of 324 V. Consequently, the available gate-current levels cannot be applied at the most effective instants as discussed in Section 4.3.1.1, as the transition completes too quickly (Fig. 4.6).

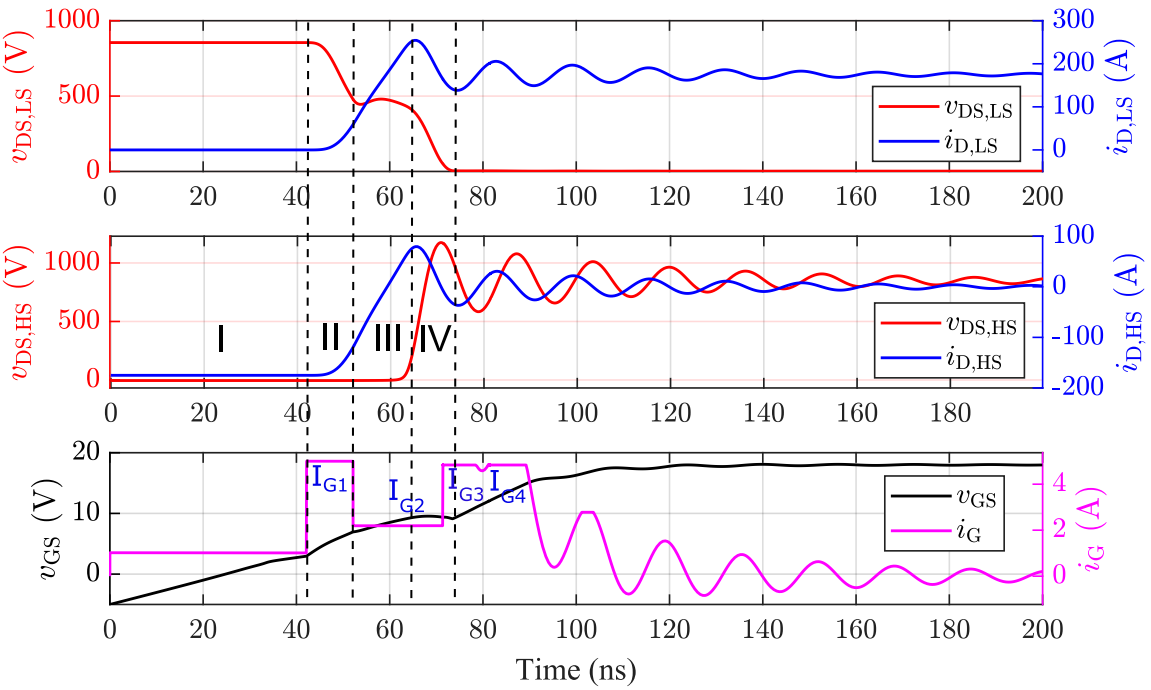


Figure 4.6: GA-optimized switching waveforms with EG120-constrained I_G profile - [9,1].

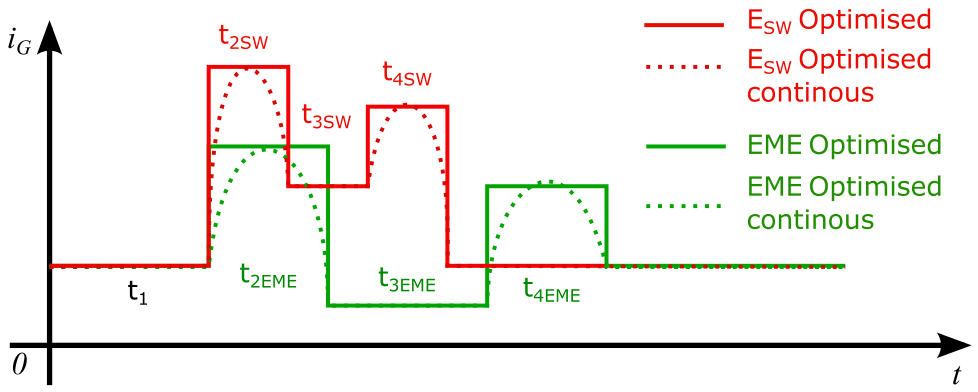


Figure 4.7: Optimised gate current profile in terms of E_{sw} and EME.

4.3.2.2 Shape of Optimized Gate-Current Waveform

After analyzing the output waveforms shown in Figs. 4.6, 4.5, and 4.3, the following conclusions can be drawn:

- The optimized gate-current waveform in both cases should always exhibit a characteristic double bell-shaped profile, which represents an effective i_G shaping strategy for MOSFET active turn-on. The corresponding switching waveforms demonstrate reduced switching loss and improved electromagnetic emission (EME) performance compared to conventional fixed-R_G operation.
- Switching loss and EME performance are strongly influenced by both the magnitude of the gate current and the duration for which it is applied within the bell-shaped profile. Fig. 4.7 illustrates the optimal i_G waveform, highlighting switching-loss-optimized and EME-optimized intervals (t_{SW} and t_{EME}), based on conclusions drawn from the observed

EG120 optimised switching waveforms. Accordingly, the gate current I_G should be applied across the different time intervals in the following manner:

  - $t < t_1$: Pre turn-on interval prior to MOSFET channel conduction, during which the exact value of I_G is not critical.
  - $t_1 < t < t_2$: A high I_G should be applied to achieve a large di_D/dt and to establish the required LS v_{DS} parasitic inductive voltage drop.
  - $t_2 < t < t_3$: A reduced I_G should be maintained to complete i_D commutation with a lower di_D/dt compared to the preceding interval.
  - $t_3 < t < t_4$: An increased I_G should be applied to accelerate voltage commutation during the Miller plateau.

- Building on these gate-current provisioning conditions, the optimization outcome in terms of switching loss or EME is governed by the applied gate current and the corresponding time intervals.
  - For the EG120 [9,1] configuration, higher gate-current levels should be applied with shorter segment durations
  - For the EG120 [1,9] configuration, lower gate-current levels should be used with longer segment durations.

  Very high i_G levels, even during the intermediate valley between the two bell-shaped peaks (corresponding to $t_{3,\mathrm{SW}}$), accelerate current commutation and reduce switching loss. However, they may induce voltage overshoot and ringing in the HS v_{DS} and LS i_D, leading to degraded EME performance (Fig. 4.7).

This optimized gate-current shape can be adapted for practical implementations by appropriately tuning the magnitude and duration of its segments according to the desired trade-off between switching loss and EME.

5 Experimental Verification and Results

5.1 Hardware Description

5.1.1 Hardware Schematic

The experimental evaluation in this work is performed using a custom-designed double-pulse test (DPT) board (Fig. 5.1). The setup enables controlled analysis of switching losses, transient

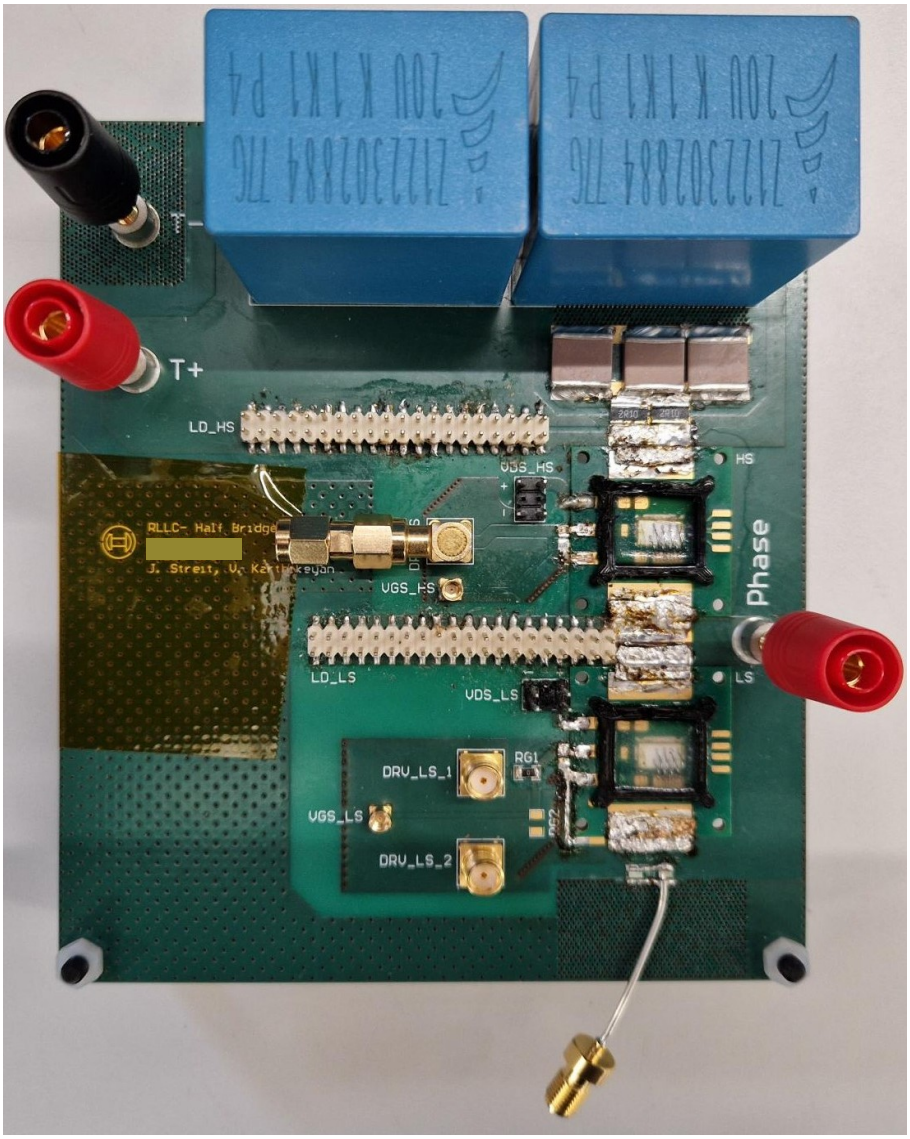


Figure 5.1: Image of the built DPT hardware.

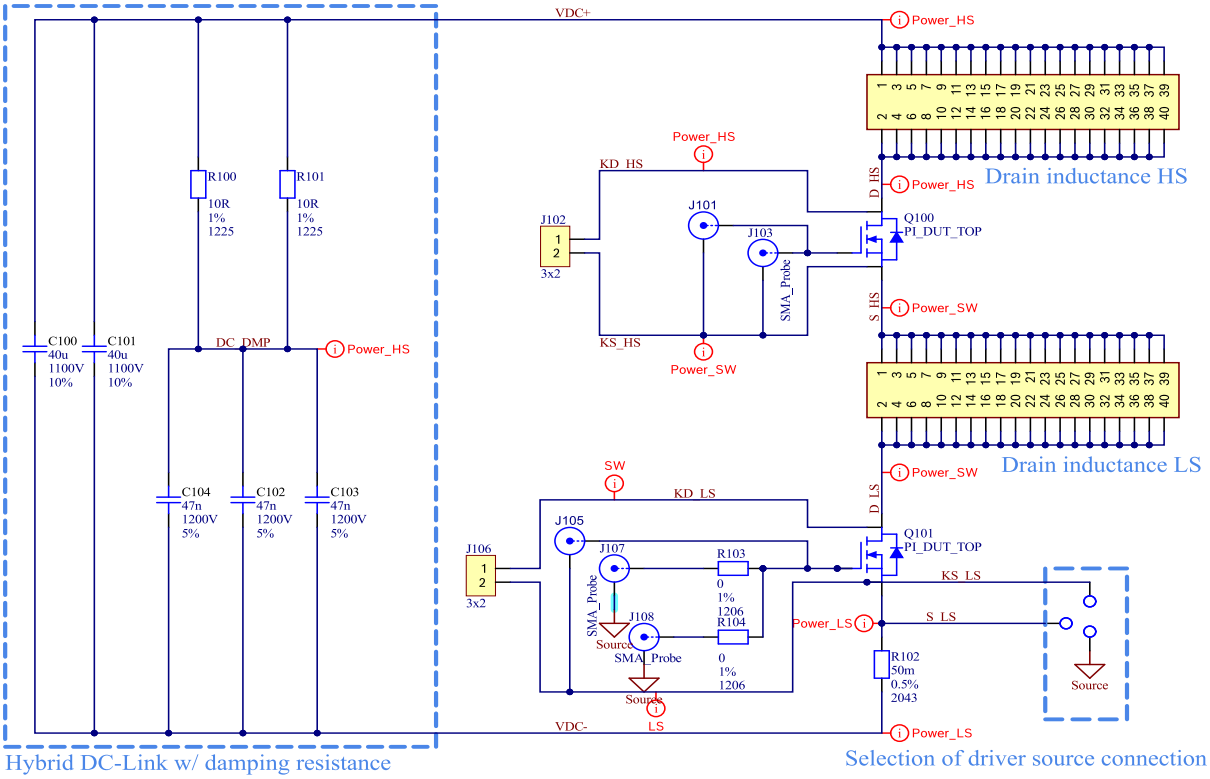


Figure 5.2: PCB Schematic of DPT Hardware.

Table 5.1: Component List

Component	**Description**	**Value (each) / Type**
Passives:		
C100, C101	DC-link capacitor	20 μF
C102–C104	Snubber capacitor	40 nF
R100, R101	Damping resistor	1 Ω
R102	Current shunt resistor of i_D	24.5 mΩ
Switches:		
Q100, Q101	SiC MOSFET (HS & LS)	BT2M1200009BOA
Interfaces:		
J101, J105	v_{GS} measurement	MMCX
J102, J106	v_{DS} measurement	MMCX
J103,	Fixed-R_G HS gate-driver input	SMA
J107	Fixed-R_G/ EG120-1 LS gate-driver input	SMA
J108	EG120-2 LS gate-driver input	SMA

phenomena, and electromagnetic emissions (EME) under defined operating conditions. The components used in the hardware setup and their respective values are listed in Table 5.1.

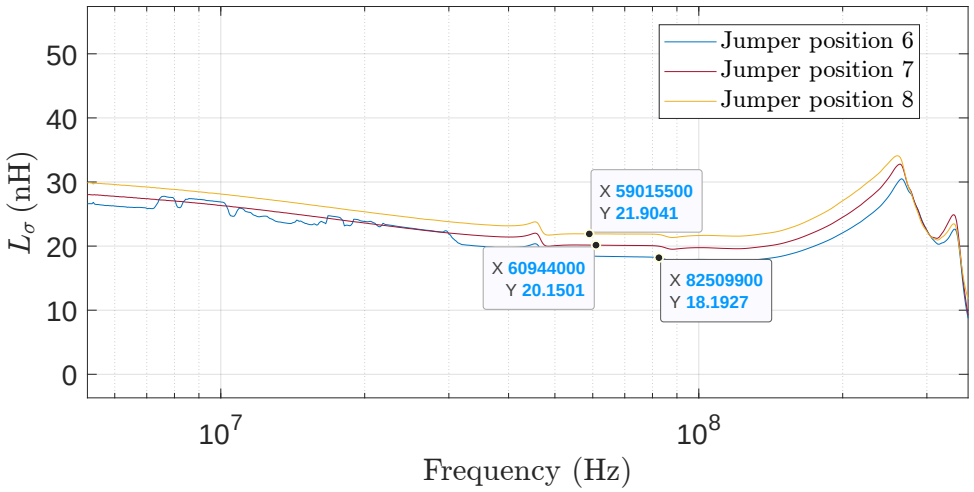


Figure 5.3: Measured L_σ for jumper positions 6, 7, and 8 corresponding to 18 nH, 20 nH, and 22 nH, respectively.

5.1.1.1 Power Semiconductor Devices

The hardware setup employs a half-bridge configuration with a HS and an LS SiC MOSFET. Both devices are Bosch SiC MOSFETs (BT2M1200009BOA), identical to those used in the state-space simulations presented in Chapters 3 and 4. The LS switch acts as the DUT.

5.1.1.2 Gate Drive and Measurement Interfaces

Both the HS and LS devices are equipped with dedicated gate drive and measurement interfaces (Fig. 5.2). High-bandwidth MMCX connectors are utilized to enable accurate measurement of gate-source voltages, drain-source voltages, and gate currents by minimizing probe loop inductance [35], while SMA connectors are utilized for gate-driving purposes. Moreover, Kelvin source and Kelvin drain connections are provided for both devices. The hardware also supports the parallel connection of two EG120 channels, which can potentially increase the number of available gate-current time intervals and enable more precise shaping of the gate current.

5.1.1.3 External Commutation Inductance

To emulate realistic conditions, external commutation inductance loops are intentionally introduced into the power loop [36] (Fig. 5.2). This inductance is implemented using configurable jumper connections, where each jumper contributes approximately 2 nH, as determined by vector network analyzer (VNA) measurements (Fig. 5.3). By selectively inserting or removing these jumpers, different loop inductance values can be realized, enabling a systematic investigation of the influence of commutation inductance on switching behavior, voltage overshoot, ringing, and EME. External inductance of up to 80 nH can be introduced by configuring the jumper connections.

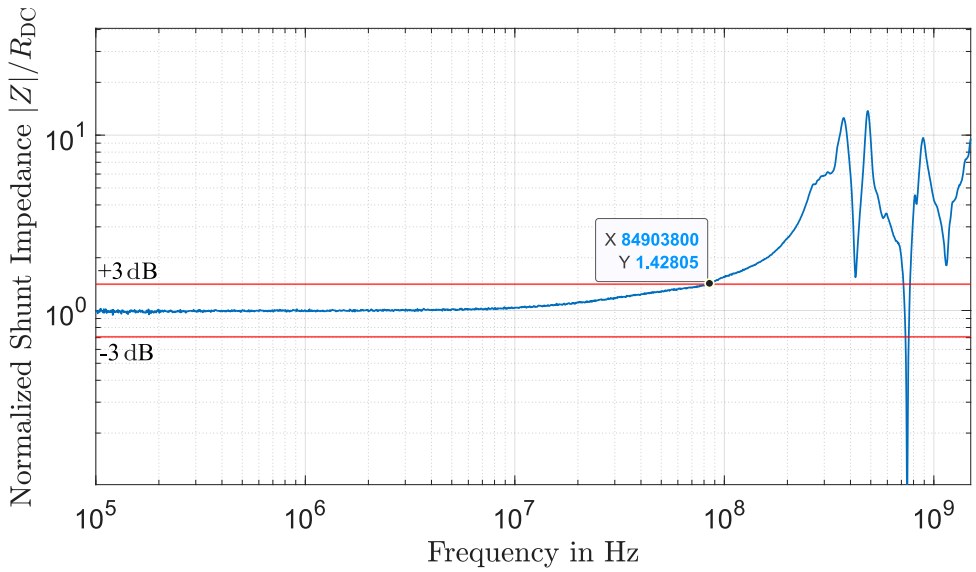


Figure 5.4: Measured frequency response of the shunt resistor.

5.1.1.4 Drain Current Measurement Using a Shunt Resistor

A low-ohmic shunt resistor is employed to measure the drain current of the LS MOSFET (Fig. 5.2). The shunt is inserted into the source path of the device. A precision shunt resistor with a resistance of 24.5 mΩ is selected as a trade-off between measurement resolution and power dissipation. The frequency response of the utilized shunt resistor is characterized using a VNA, as shown in Fig. 5.4. The transfer function exhibits a relatively flat magnitude response up to 10 MHz and remains within the 3 dB bandwidth up to approximately 85 MHz, as indicated by the red bounds in Fig. 5.4. Within this bandwidth, the resistive behavior dominates, ensuring accurate reconstruction of the drain current without significant amplitude distortion.

5.1.2 PCB Design

A four-layer PCB is employed in this thesis to achieve low parasitic inductance and a well-controlled commutation loop. The PCB stack-up is optimized to separate the power, gate-drive, and measurement paths while maintaining compact current loops.

A vertical layout concept is adopted for the half-bridge power loop, with the forward and return current paths arranged on opposing PCB layers (Fig. 5.5). This configuration minimizes the effective loop area compared to conventional horizontal layouts. The opposing currents on adjacent layers provide partial magnetic field cancellation, thereby further reducing the commutation loop inductance [32]. This layout is well suited for high-speed SiC MOSFET switching, as reduced loop inductance helps suppress voltage overshoot, ringing, and high-frequency emissions.

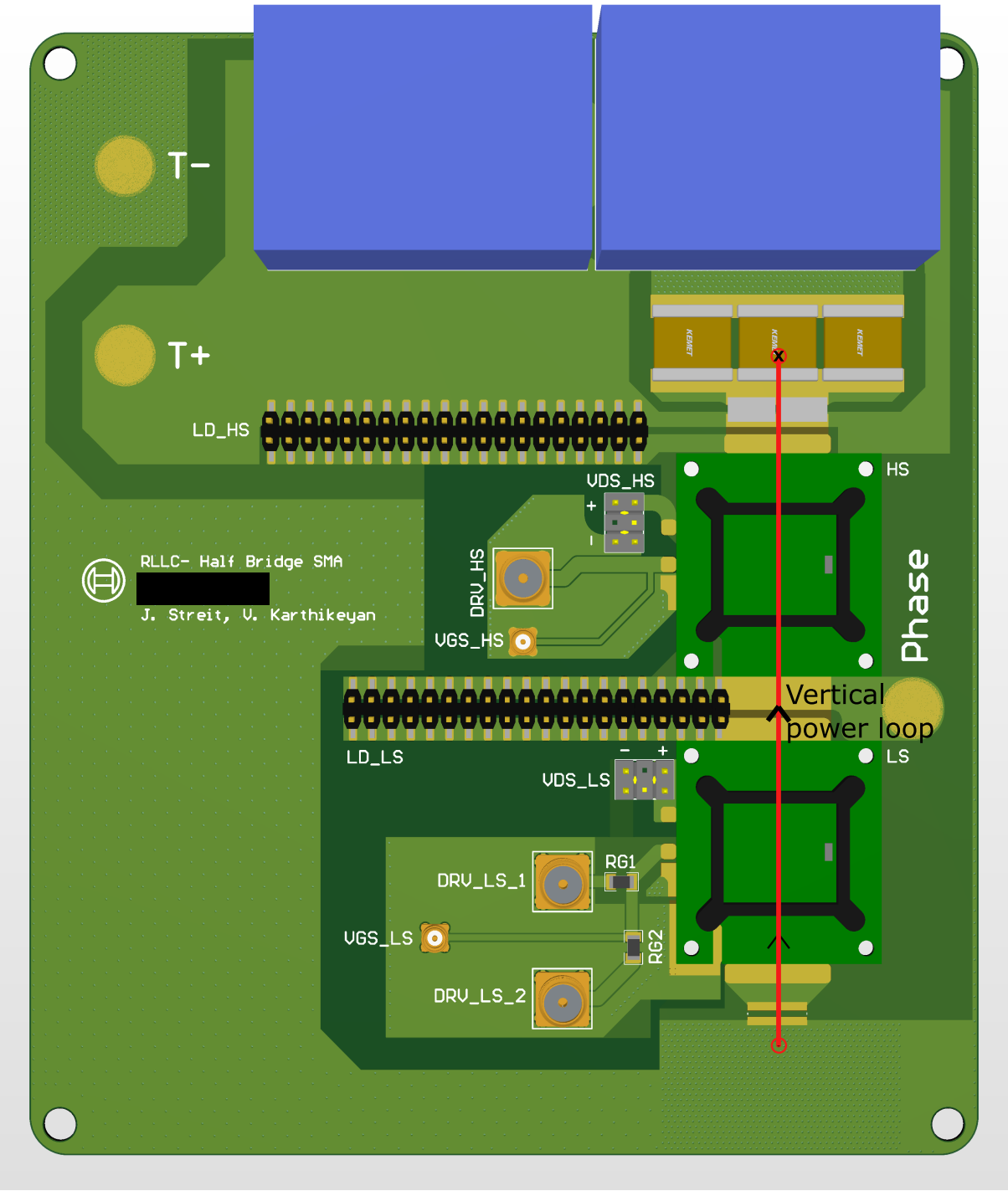


Figure 5.5: PCB Design of DPT Hardware.

5.2 Experiment Setup and Test Procedure

The experimental validation is carried out using a DPT configuration (Fig. 5.6). Fig. 5.7 presents a schematic representation of the complete experimental setup, detailing the current and voltage measurement points. Table 5.2 summarizes the measurement techniques employed in the experimental setup.

An inductive load is connected between the switching node and the DC-link to establish a controlled load current during the test. Gate-drive signals are generated using an external function generator, which provides the pulse sequence required for the DPT (Fig. 2.1). The HS device is driven through a fixed gate resistance driver and is kept permanently off during the measurements. The LS device, which acts as the DUT, is driven either through a fixed gate resistance driver or the EG120 gate driver, depending on the specific test condition. Both gate drivers are supplied from isolated low-voltage sources.

5.2.1 Test Procedure

All experiments were conducted at a fixed operating point summarized in Table 5.3. Four distinct test cases were investigated, each representing a different gate-driving strategy for the LS MOSFET during the turn-on transition.

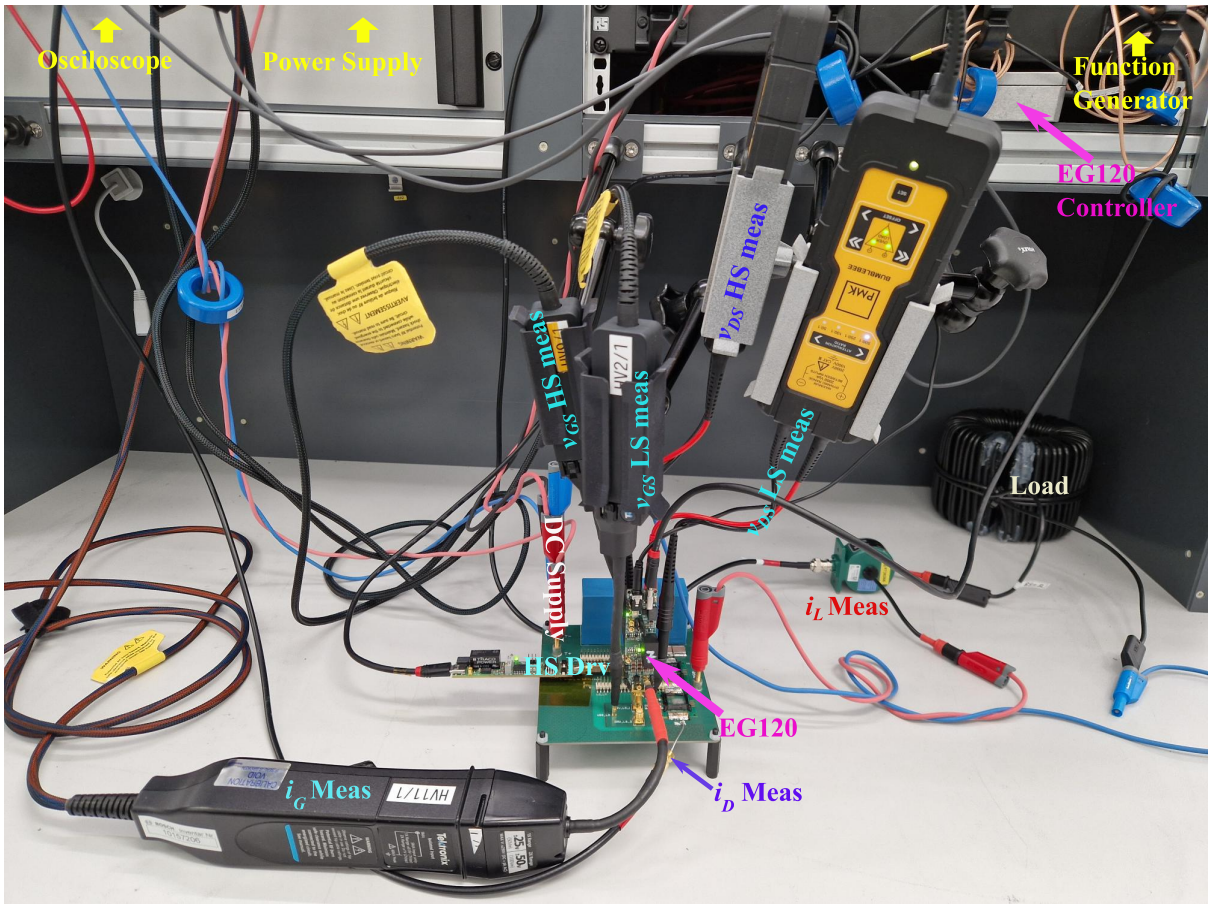


Figure 5.6: Experiment setup.

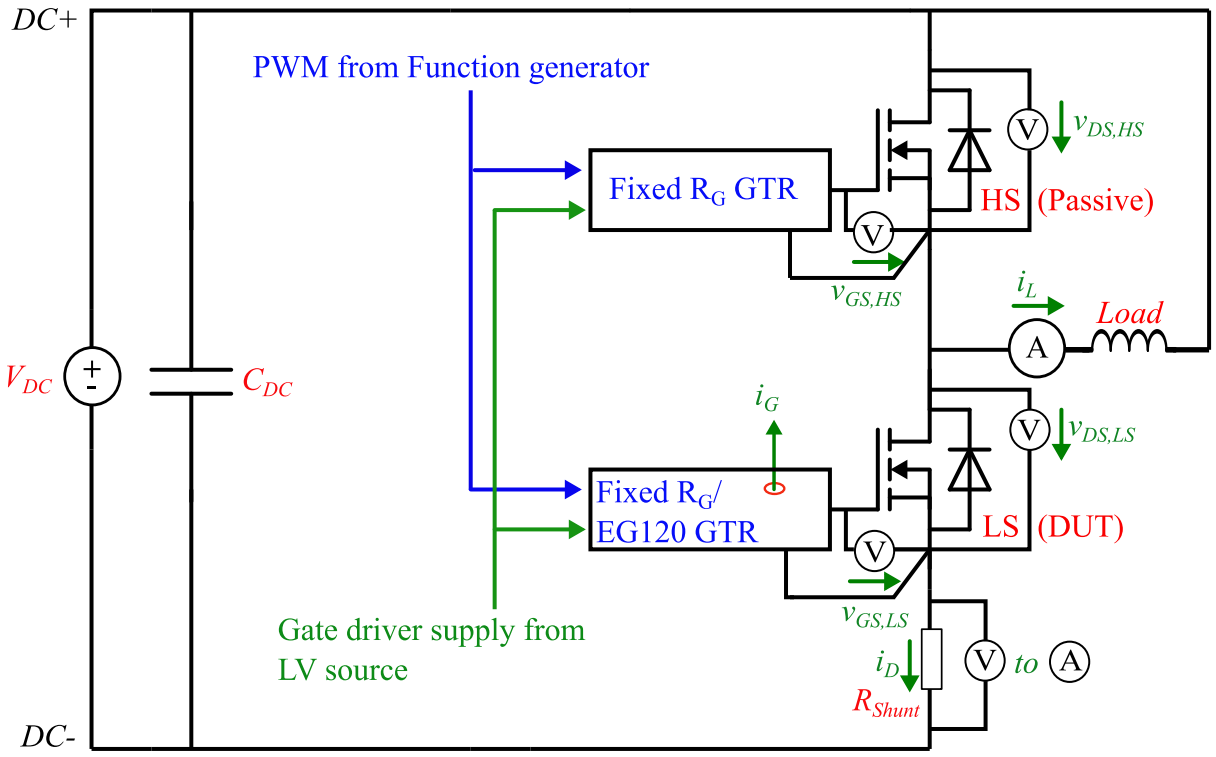


Figure 5.7: Experiment setup schematic.

Table 5.2: Measurement Methods

Quantity	**Measurement Method**
$v_{GS,\mathrm{LS}}$	IsoVu probe
$v_{GS,\mathrm{HS}}$	IsoVu probe
i_D	Passive shunt resistor
i_L	Pearson current probe
i_G	IsoVu probe
$v_{DS,\mathrm{LS}}$	Bumblebee differential probe
$v_{DS,\mathrm{HS}}$	Bumblebee differential probe

- Fixed gate-resistor driver with $R_G = 5.1\,\Omega$.
- Fixed gate-resistor driver with $R_G = 7.5\,\Omega$.
- EG120 gate driver employing the gate-current profile obtained via a genetic algorithm optimized for minimum switching loss ([9,1]).
- EG120 gate driver employing the gate-current profile obtained via a genetic algorithm optimized for minimum electromagnetic emissions (EME) ([1,9]).

Table 5.3: Circuit Parameters and Operating Conditions

Symbol	Description	Value
L_σ	Commutation loop inductance	20 nH
V_{DC}	DC-link voltage	850 V
I_{Load}	Load current	175 A
L_{Load}	Load inductance	56 μH
$V_{\mathrm{GS,high}}$	Gate–source voltage (turn-on)	18 V
$V_{\mathrm{GS,low}}$	Gate–source voltage (turn-off)	−5 V

5.3 Results and Analysis

For each test case, the LS MOSFET turn-on transient was recorded and analyzed. During the experimental evaluation, the hardware experienced peak drain currents up to 275 A and peak drain–source voltages up to 1000 V. Different driving strategies were compared based on two key performance metrics: *Switching loss* and *EME*. Tables 5.4 and 5.5 compare the switching loss and EME performance of four different gate-driving strategies, with fixed $R_G = 5.1\,\Omega$ used as the reference case. The switching loss and EME metric (HGI) are calculated as per Section 3.1.3.

5.3.1 Fixed Gate-Resistance Driver Tests

Figures 5.8 and 5.9 illustrate the active turn-on transition waveforms obtained with fixed gate resistances of 5.1 Ω and 7.5 Ω, respectively. Similar to the simulation results, the 5.1 Ω gate resistance case results in lower switching losses compared to the 7.5 Ω case but suffers increased voltage overshoots and ringing, which is reflected as higher HGI values (Tables 5.4 and 5.5). Conversely, operation with a 7.5 Ω gate resistance leads to reduced overshoot and ringing at the expense of higher switching losses.

5.3.2 EG120 Driver Tests

Figures 5.10 and 5.11 illustrate the active turn-on switching waveforms obtained using the EG120 gate driver with gate-current profiles optimized for reduced EME ([1,9]) and reduced switching loss ([9,1]), respectively. The corresponding gate-current profiles programmed in the

Table 5.4: Comparison of t_{on} and E_{sw} for different cases, referenced to fixed $R_G = 5.1\,\Omega$, at the operating point as per Table 5.3

Method	t_{on} (ns)	E_{sw} (μJ)	ΔE_{sw} (%)
Fixed $R_G = 5.1\,\Omega$	43.98	2580.3	–
Fixed $R_G = 7.5\,\Omega$	57.42	3309.8	+28.3
EG120 [9,1] (Loss opt.)	41.12	**2209.0**	**−14.4**
EG120 [1,9] (EME opt.)	53.60	3775.3	+46.3

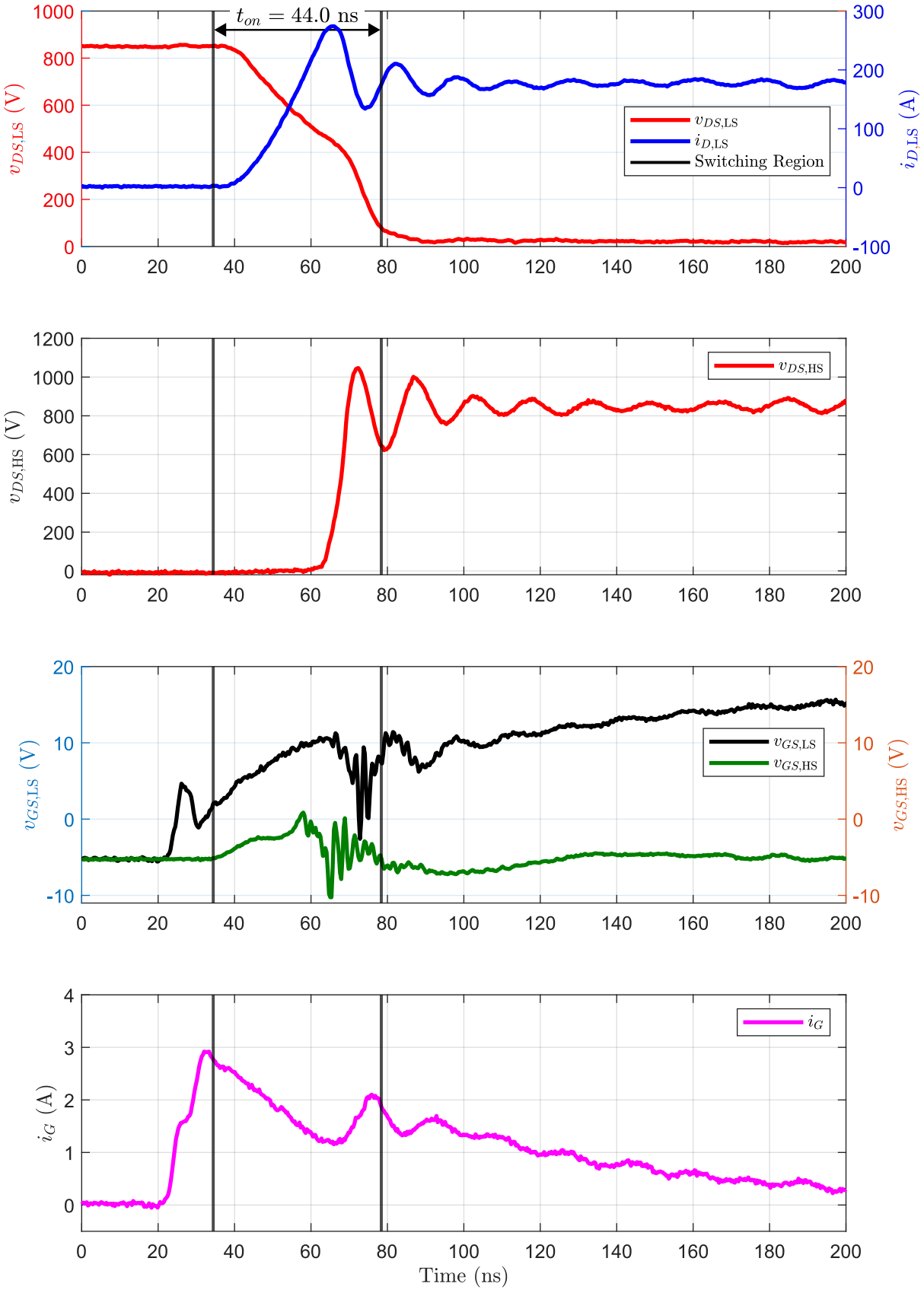


Figure 5.8: Measured switching waveforms for fixed gate-resistance operation with R_G = 5.1 Ω.

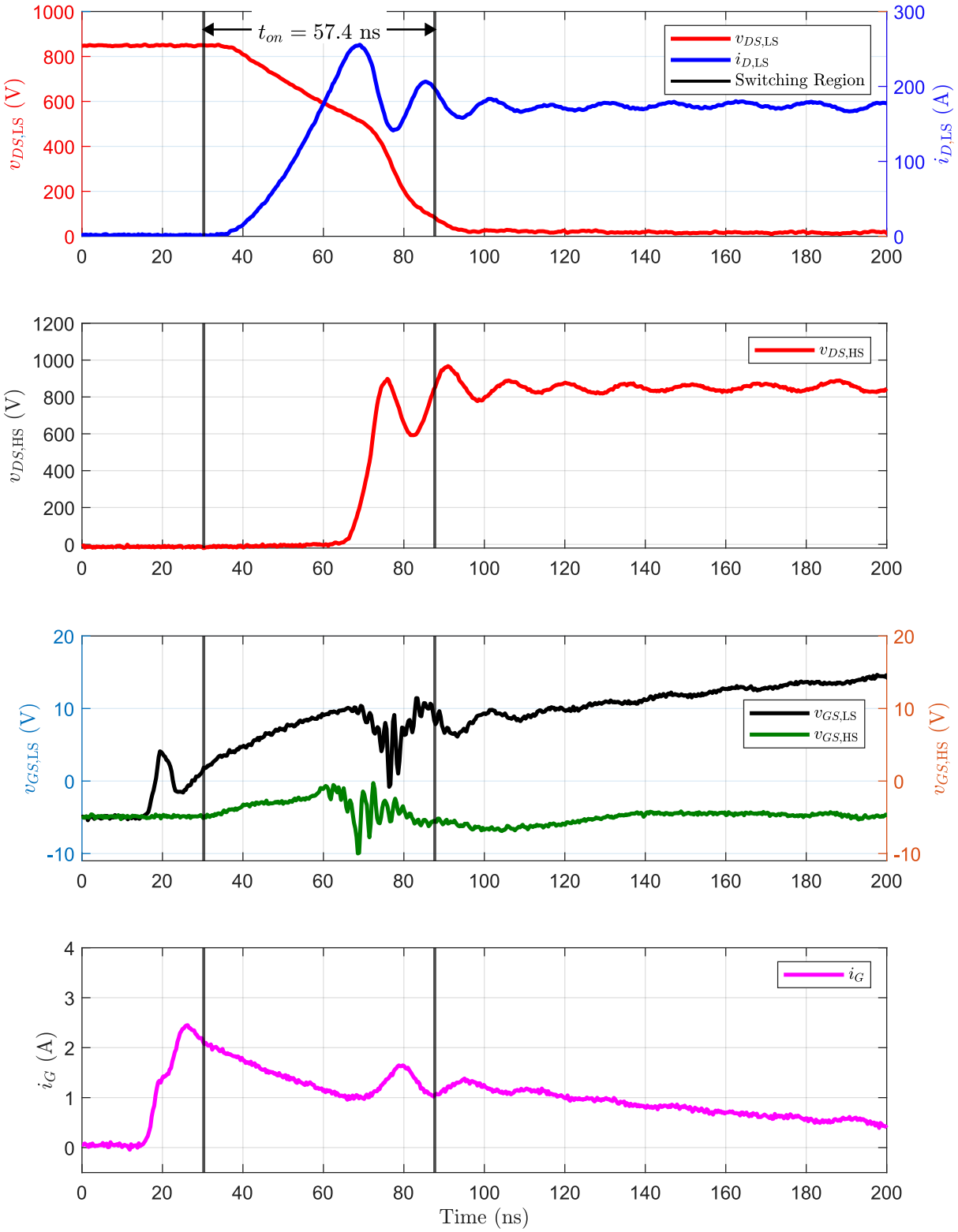


Figure 5.9: Measured switching waveforms for fixed gate-resistance operation with R_G = 7.5 Ω.

Table 5.5: Comparison of HGI and $V_{DS\,HS,OS}$ for different cases, referenced to fixed $R_G = 5.1\,\Omega$, at the operating point as per Table 5.3

Method	HGI_{LS}	HGI_{HS}	ΔHGI_{HS} (%)	$V_{DS\,HS,OS}$ (V)	$\Delta V_{DS\,HS,OS}$ (%)
Fixed $R_G = 5.1\,\Omega$	22.47	25.54	–	196	–
Fixed $R_G = 7.5\,\Omega$	13.12	18.85	−26.2	117	−40.3
EG120 [9,1] (Loss opt.)	15.03	**19.99**	**−21.7**	**103**	**−47.4**
EG120 [1,9] (EME opt.)	10.16	**10.58**	**−58.6**	**79**	**−59.7**

EG120 are summarized in Table 5.7. Consistent with the simulation results, the EG120 [9,1] configuration achieves the lowest switching loss, while the EG120 [1,9] configuration yields the lowest HGI values among all considered cases (Tables 5.4 and 5.5).

5.3.3 Overall Performance Improvement Using EG120-Optimized Profiles

The EG120 configuration optimized for reduced switching loss ([9,1]) (Fig. 5.11) yields the best overall switching performance by achieving the lowest switching loss of 2209 μJ. This corresponds to a relative switching-loss reduction of 14.4% compared to the 5.1 Ω fixed gate-resistance case, which previously exhibited the lowest switching loss among the fixed-resistance configurations, while simultaneously achieving a 21.7% lower HGI_{HS} value, indicating improved EME performance (Tables 5.4 and 5.5). Consequently, this configuration demonstrates superior performance in terms of both switching loss and EME metrics. When compared to the 7.5 Ω fixed gate-resistance case, the same EG120 configuration provides a 42.6% reduction in relative switching loss while maintaining a comparable EME performance, with only a marginal degradation (Tables 5.4 and 5.5).

In contrast, the EG120 configuration optimized for EME reduction ([1,9]) exhibits significantly reduced voltage overshoot and ringing relative to the other configurations, resulting in a HS v_{DS} waveform that more closely resembles the rising edge of a Gaussian-shaped transition (Fig. 5.10). This behavior leads to the lowest HGI values among all configurations, with HGI_{LS} = 10.16 and HGI_{HS} = 10.58 (Table 5.5). However, as expected, it is accompanied by higher switching losses.

5.3.4 Improvement in Switching Loss

The reduction in switching loss is primarily achieved through a shortened switching duration in the EG120 [9,1] configuration. Meanwhile, the EG120 [1,9] configuration, although exhibiting a slightly faster transition, results in higher switching losses than the 7.5 Ω case (Table 5.4). This is attributed to the increase in the v_{DS} waveform towards the end of Phase III due to the intentional reduction of $\mathrm{d}i_D/\mathrm{d}t$ to very low values in order to obtain the best possible waveform in terms of EME performance.

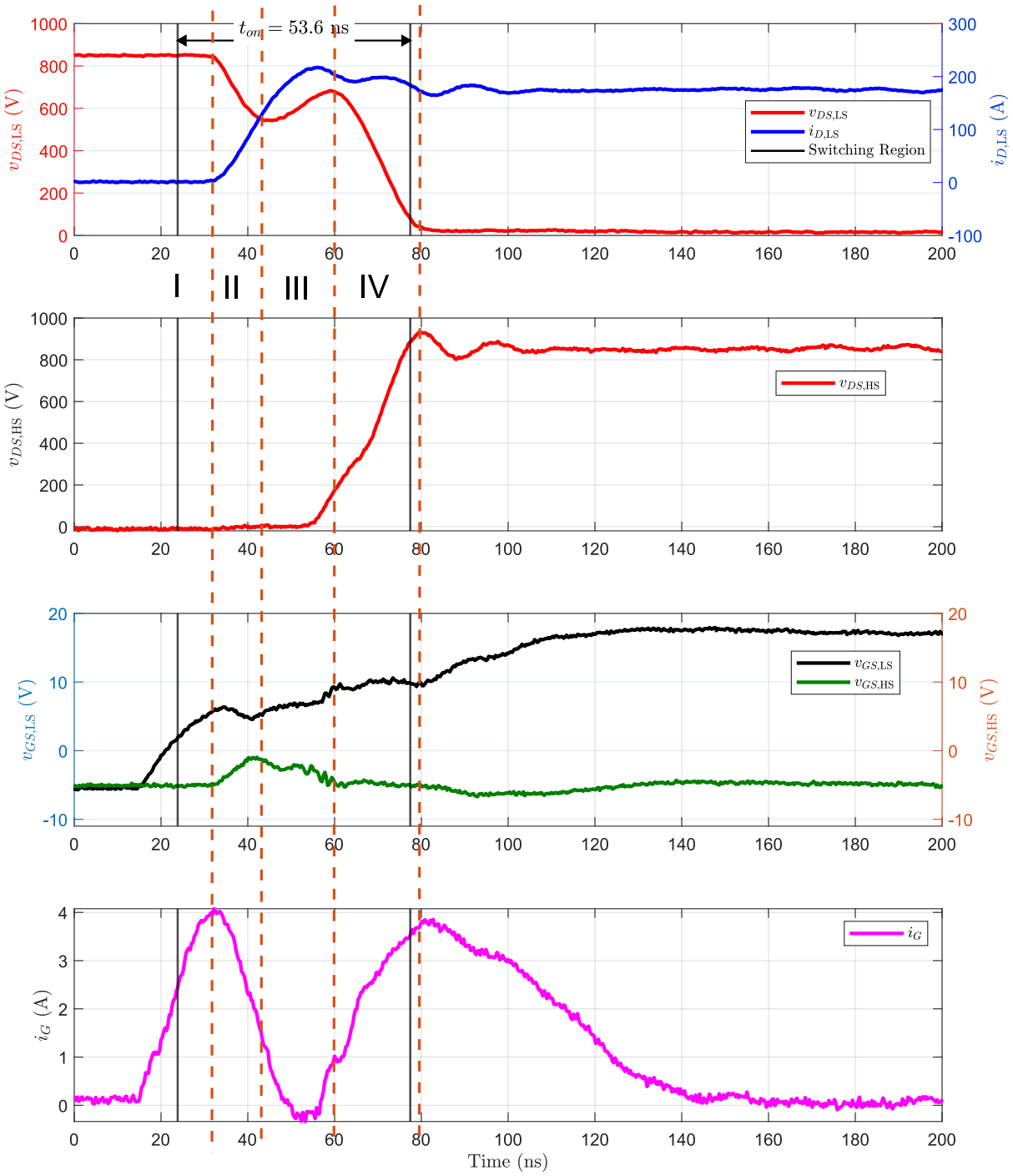


Figure 5.10: Measured switching waveforms for EG120 [1,9] configuration.

Table 5.6: Comparison of HS switching transition characteristics and EME-related metrics

Method	$v_{DS,HS}$ t_{10-90} (ns)	σ_t (ns)	σ_ω (MHz)	f_{cut2} (MHz)	HGI_{HS}
Fixed R_G = 5.1 Ω	4.8	68.5	37.3	53.50	25.54
Fixed R_G = 7.5 Ω	6.08	54.5	34.6	41.88	18.85
EG120 [9,1]	5.76	60.0	33.3	44.21	19.99
EG120 [1,9]	17.28	59.1	17.9	14.74	10.58

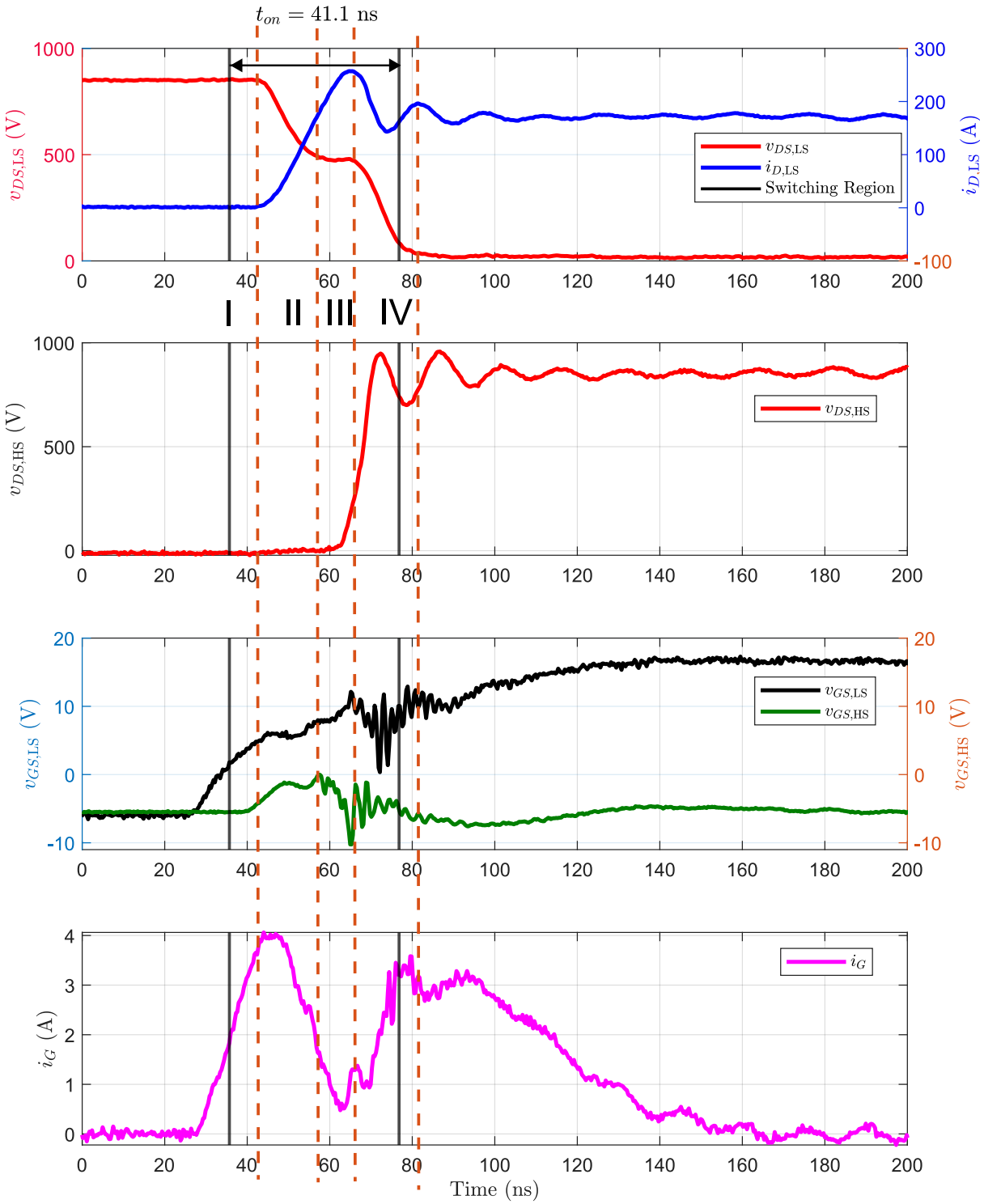


Figure 5.11: Measured switching waveforms for EG120 [9,1] configuration.

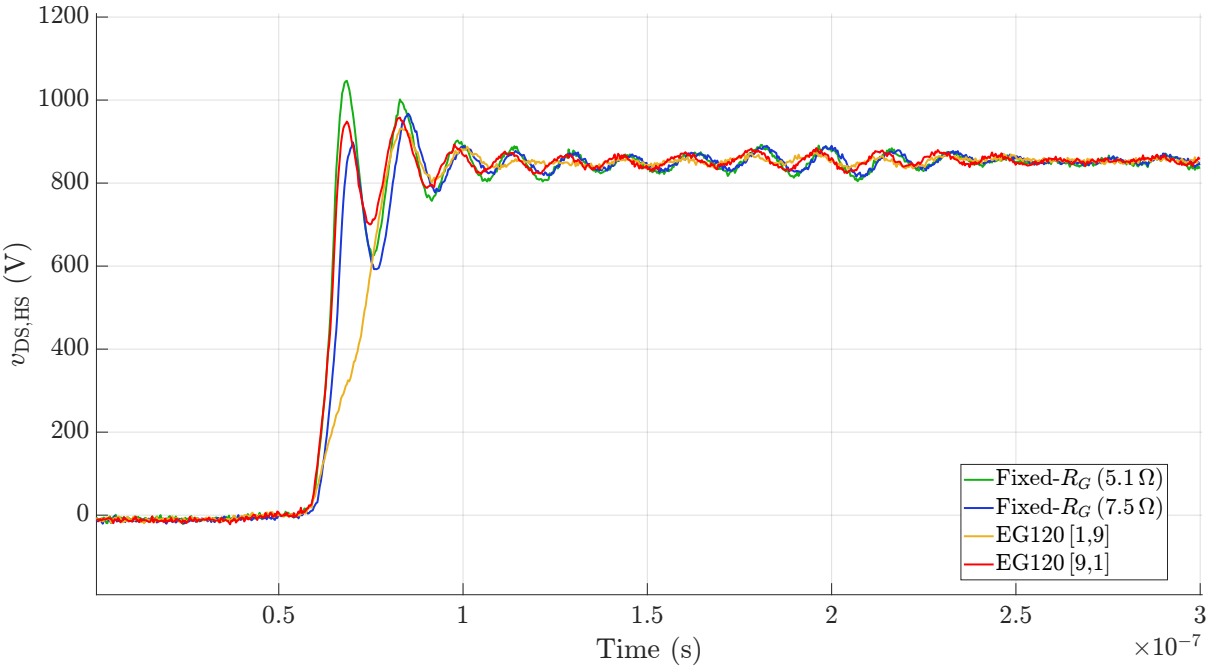


Figure 5.12: Comparison of overshoot and ringing in HS v_{DS} waveforms.

5.3.5 Improvement in EME Performance

Table 5.6 presents a detailed comparison of the parameters related to EME performance for all considered cases. These values are calculated according to (2.10), (3.14), and (3.17), using the parameters extracted from the switching waveforms.

5.3.5.1 Temporal dispersion

Figure 5.12 compares the HS v_{DS} waveforms for all considered cases. As shown in Table 5.6, a faster transition does not necessarily correspond to lower temporal dispersion, as ringing and voltage overshoot also contribute significantly in addition to the transition time. Consequently, the EG120 [1,9] case exhibits a small σ_t despite a very high $t_{10\text{–}90}$, owing to its minimal ringing and overshoot.

5.3.5.2 Frequential dispersion

Figure 5.13 compares the HS v_{DS} waveforms for all considered cases in the frequency domain after Fourier transform. Both techniques introduced in Section 2.3.3.2 are employed to reduce the frequency dispersion of the switching transients.

- Owing to the longer transition period in the EG120 [1,9] case, the resulting f_{cut2} is significantly lower, which leads to a substantial reduction in frequency dispersion in this case.
- The EG120 [1,9] case also exhibits a much steeper decay beyond the cutoff frequency. This decay is not linear but curved, allowing the spectrum to reach the noise floor more

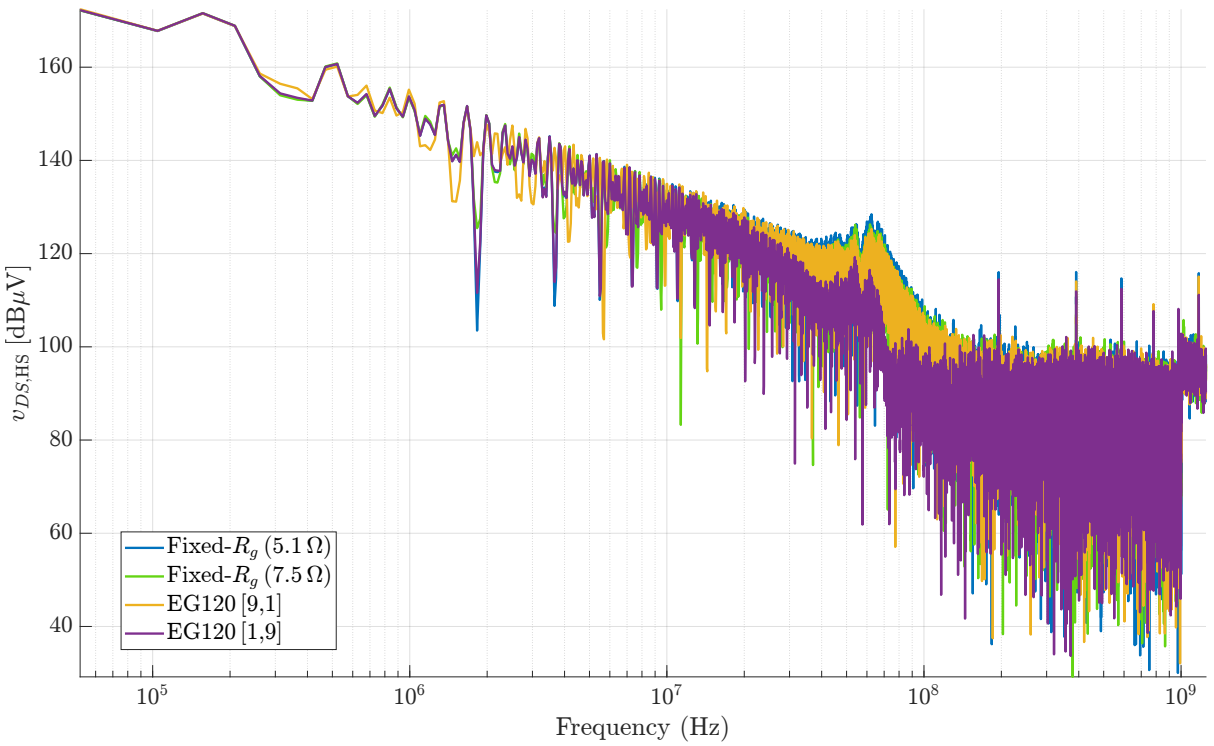


Figure 5.13: Comparison of frequency spectra in different HS v_{DS} waveforms.

Table 5.7: EG120 gate-current levels used in lab measurements

Configuration	I_{G1} (A)	I_{G2} (A)	I_{G3} (A)	I_{G4} (A)	I_{G5} (A)	I_{G6} (A)
EG120 [1:9] I_G profile	8.00	1.847	6.55	1.345	1.345	1.345
EG120 [9:1] I_G profile	9.00	2.70	7.00	2.78	2.78	2.78

rapidly (Figure 5.13). This behavior is attributed to the HS v_{DS} waveform exhibiting a transition that closely resembles the rising edge of a Gaussian waveform.

5.3.5.3 Limitations of EG120

Tables 5.7 and 5.8 present the EG120 gate-current and timing values programmed for both cases, which aim to replicate the simulation profiles under the constraints imposed by the EG120 hardware (Fig. 4.2). From Figures 5.10 and 5.11, it can be observed that the bell-shaped gate-current profiles do not occur at the exact positions or phases discussed in Section 4.3.2.2 (Fig. 4.5). Furthermore, both the gate-current amplitudes and the corresponding timing parameters vary significantly compared to the values used in the simulation (Table 4.5), and the distribution across individual segments also differs noticeably.

Table 5.8: EG120 gate-current timing parameters used in lab measurements

Configuration	t_1 (ns)	t_2 (ns)	t_3 (ns)	t_4 (ns)	t_5 (ns)
EG120 [1:9] I_G profile	14.4	20.0	20.8	9.98	147.0
EG120 [9:1] I_G profile	12.0	15.00	25.10	10.0	147.0

The discrepancies between the simulated and experimental gate-current profiles can be attributed to the following practical limitations:

- The EG120 gate driver requires a finite gate current rise time when transitioning between programmed current levels. Therefore, it cannot reproduce the ideal rectangular transitions assumed in the simulation.
- To achieve the required effective gate-current levels in practice, higher programmed current values are necessary in each segment. For example, in the EG120 [1,9] configuration, a programmed current of 8 A is required in the first segment to attain an effective peak gate current of approximately 4 A in the measured waveform.
- The programmed timing parameters do not directly translate to the observed experimental timing due to internal delays and dynamic effects within the driver and the setup. As a result, both the timing and current levels must be manually adjusted during experimentation to approximate the simulated profile, thereby resulting in deviations from the simulated current and timing profiles.

As a result, higher I_G levels are inevitably applied during certain intervals where lower values would be preferable, and vice versa, relative to the ideal waveform (Fig. 4.7), reducing the overall achievable performance in both cases.

Nevertheless, the obtained waveforms can be classified as optimized for switching loss and EME, as they are obtained from deliberately selected I_G levels and corresponding time intervals. The EG120 [9,1] configuration exhibits a very short overall switching transition, with higher I_G levels applied in each segment and shorter durations for each level compared to the EG120 [1,9] case. For the EG120 [9,1] configuration, I_G drops only to a minimum value of approximately 0.6 A, whereas for the EG120 [1,9] configuration it briefly drops slightly below zero (Figures 5.10 and 5.11). Although the absolute I_G values are not replicated from the simulation, the characteristic behavior of the waveform optimized for switching loss and EME, as discussed in Section 4.3.2.2, is successfully reproduced.

Despite these limitations and the inability to achieve optimal simulated performance, the EG120-based approach yields significantly improved results compared to the conventional fixed gate-resistance method, indicating that a closer approximation of the ideal gate-current profile could yield even greater performance gains.

6 Summary and Conclusion

6.1 Summary

This thesis successfully demonstrated the experimental validation of a gate-current optimization framework for SiC MOSFETs, enabling a significantly reduced trade-off between switching losses (E_{sw}) and electromagnetic emissions (EME) compared to conventional fixed gate-resistance driving. Each of the four major research questions covering the definition of optimal switching behavior, the application of genetic algorithms (GA) for gate-current optimization, the experimental hardware implementation of the optimized gate-current profiles, and their associated limitations were comprehensively investigated, and successful results were obtained.

A physics-based state-space simulation framework was introduced and calibrated using experimentally extracted device parameters, including nonlinear device capacitances (C_{GS}, C_{GD}, C_{DS}), drain-current (i_D), and loop inductance (L_σ), which was subsequently employed to perform double-pulse test (DPT) simulations of the chosen Bosch SiC MOSFET. The fixed gate-resistance simulation results were then visually compared with experimental DPT measurements to verify that the simulation framework sufficiently captures the real-world switching behavior and can therefore be used for the subsequent optimization study.

A GA–based optimization framework was then used to derive optimal gate-current profiles. Two distinct optimization methodologies were investigated: a continuous gate-resistance optimization approach used for theoretical benchmarking, and a discrete gate-current optimization approach that explicitly incorporates some practical constraints of the EG120 programmable gate driver, including limited gate current amplitude and discrete time intervals. This optimization employed a multi-objective cost function that quantitatively balances switching loss and EME (quantified using the Heisenberg–Gabor Inequality (HGI)), while ensuring drain-source voltage overshoot constraints.

Simulations employing the optimized gate-current profiles demonstrated clear performance improvements compared to conventional fixed gate-resistance operation. In this thesis, E_{sw} is evaluated for the LS MOSFET during turn-on, while EME is assessed based on the v_{DS} waveforms of both the LS and HS devices. For the EME-optimized case ([1,9]), the HS v_{DS} exhibited a waveform shape closer to a Gaussian function with negligible overshoot and ringing, resulting in an $\mathrm{HGI}_{\mathrm{HS}}$ value of 0.53, corresponding to a 91.5% reduction compared to the fixed $R_G = 5.1\,\Omega$ case, while also achieving a 12% reduction in switching loss. In contrast, the switching-loss-optimized case ([9,1]) achieved the lowest switching loss among all evaluated scenarios (1396.7 μJ), corresponding to a 50.2% reduction compared to the fixed $R_G = 5.1\,\Omega$ case, but at the expense of degraded EME performance.

The theoretically derived gate-current profiles were applied using the EG120 current-source gate driver on a custom-designed double-pulse test (DPT) hardware. During the experiment, the hardware experienced peak drain currents of up to 275 A and peak drain–source voltages

of up to 1000 V, with the option to introduce external commutation inductance of up to 80 nH via configurable jumper connections. At the operating point of L_σ = 20 nH, V_{DC} = 850 V, and I_{Load} = 175 A, the EG120 configuration optimized for switching-loss reduction ([9,1]) achieved a minimal E_{sw} of 2209 μJ, corresponding to a relative reduction of 14.4% compared to the fixed gate-resistance case with the lowest switching loss (R_G = 5.1 Ω). Moreover, it also exhibited reduced EME metrics, with a 21.7% reduction in $\mathrm{HGI}_{\mathrm{HS}}$ and a 47.4% reduction in the V_{DS} voltage overshoot. When benchmarked against the 7.5 Ω fixed R_G case, the same EG120 configuration yielded a switching-loss reduction of 42.6% while maintaining a comparable EME performance, with only marginal degradation.

Moreover, the EME-optimized EG120 configuration ([1,9]) achieved the best overall EME performance among all investigated experiment cases, with $\mathrm{HGI}_{\mathrm{LS}}$ = 10.16 and $\mathrm{HGI}_{\mathrm{HS}}$ = 10.58. This corresponds to a 58.6% reduction in $\mathrm{HGI}_{\mathrm{HS}}$ and a 59.7% reduction in the V_{DS} overshoot compared to the 5.1 Ω fixed R_G configuration. These experimental results confirmed that the implemented EG120 gate-current profiles successfully reproduced the intended optimization objectives, despite practical hardware constraints.

Observed discrepancies between simulated and measured results were systematically analyzed and were attributed to the finite current rise times, internal delays, and limited maximum current-providing capability of the EG120 driver, which prevented the exact reproduction of the ideal simulated gate-current profiles. Despite these hardware limitations, the results demonstrated that optimized gate-current driving significantly outperforms conventional fixed R_G operation, effectively reducing the traditional efficiency–EME trade-off in modern power electronic converters.

6.2 Conclusion

The experimental validation in this work was conducted with L_σ of approximately 20 nH. In industrial power electronic systems employing complex converter topologies, such as multilevel inverters, more intricate busbar structures inevitably result in significantly higher L_σ [37]. In such high-inductance regimes, voltage overshoot becomes the primary limitation on switching speed, leading to increased switching losses. Under these conditions, the optimized gate-current profiling method employed in this thesis enables selective shaping of the switching transient, rather than uniformly slowing the entire commutation process. This indicates that the performance advantage of the gate-driving method discussed in this thesis becomes even more pronounced in high-inductance power electronic systems.

While theoretically optimal gate-current profiles can substantially reduce switching losses and EME, their practical realization is often constrained by the limitations of commercially available current-source gate drivers, such as the EG120. Future research may therefore focus on bridging the gap between ideal continuous-resistance profiles or current-source gate-driving concepts and practical voltage-source implementations, thereby avoiding the need for dedicated current-source drivers. Discrete analog closed-loop approaches, such as using an auxiliary MOSFET in parallel with the gate resistor [38] or external analog control circuits based on differentiators and high-speed comparators with $\mathrm{d}v_{DS}/\mathrm{d}t$ feedback [8], can enable quasi-continuous gate-current or gate-resistance profiles that approximate towards the theoretical optimum. However, the increased circuit complexity of such approaches may limit their suitability for cost and reliability-driven industrial applications.

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